An Introduction to Higher Algebraic K-Theory

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Abstract

Algebraic K-theory associates to a ring R a sequence of abelian groups $K_n(R)$ encoding its finitely generated projective modules, in analogy with the topological K-theory of vector bundles. This thesis gives an introduction to the subject, from the classical low-degree groups to Quillen's definition of the higher groups.

We begin with projective modules and Swan's theorem, then construct the Grothendieck group K_0 for rings, symmetric monoidal categories and exact categories. Next we study the classical groups $K_1(R) = GL(R)/E(R)$ and $K_2(R)$.

After reviewing the required homotopy theory, including classifying spaces of categories, H-spaces, Quillen's Theorems A and B, and the +-construction, we define the higher groups $K_n(R)$ in three ways: through the $BGL(R)^+$ construction, the $S^{-1}S$ construction, and Quillen's Q-construction. We show that these agree with the classical groups in low degrees and with each other, culminating in the "$+ = Q$" theorem.

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Chapter 1

Preliminaries

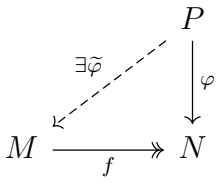
The subject of K-Theory was historically concerned with studying vector bundles on spaces. In Algbraic K-Theory, the primary objects of interest are projective modules over a ring, and vector bundles over schemes.

1.1 Projective Modules

Let R be a (unital) associative ring. We mention the standard notion of projective modules and illustrate its relevance to the subject of K-theory.

Projective modules are closely related to free modules, and in fact.

Definition 1.1 (Projective Module)**.** We say P is a projective (right[1]) R-module if it satisfies the following property: Let $M \to N$ be a surjection of (right) R-modules, then any map $\varphi : P \to N$ lifts to a map $\widetilde{\varphi} : P \to M$ such that the following diagram commutes:



It is a standard result that this definition is equivalent to the various following criteria:

Proposition 1.2. Let R be as before and P be an (right) R-module. Then the following are equivalent:

(i) P is projective.

[1] we fix a convention, either right or left

(ii) For any surjection $M \twoheadrightarrow N$ of (right) R-modules, the induced map $\mathrm{Hom}(P, M) \to \mathrm{Hom}(P, N)$ is a surjection.
(iii) Any surjection $\pi : M \twoheadrightarrow P$ of (right) R-modules splits, i.e., there is $s : P \to M$ such that $\pi \circ s = 1_P$.
(iv) There exists an R-module Q such that $P \oplus Q$ is a free R-module.

1.1.1 Basic Properties of Projective Modules

When the ring R is commutative, right R-modules can simply be called R-modules.

Example 1.1.1. All free modules are obviously projective.

We claim that there exist non-free projective modules, and the failure of all projective modules to be free shall be a major core of the theory henceforth.

Example 1.1.2. Let R and S be non-trivial commutative rings. Then $R \oplus 0$ and $0 \oplus S$ are unsurprisingly projective $R \oplus S$-modules since $R \oplus S$ is a free $R \oplus S$ module. Yet, neither is free.

It is a question of interest to characterize when are projectives are free.

Proposition 1.3. If R is a field or a division ring, then all (right) projective modules over R are free.

We also have the following result for commutative rings R:

Proposition 1.4. All projective modules over a (commutative) local ring R are free.

It is a standard exercise in algebra to show this is true for finitely generated projective modules. However, the result holds more generally for any projective module over R which we refer the reader to [Kaplansky].

Proposition 1.5. If $\mathfrak{p}$ is a prime ideal of a commutative ring R and P is a finitely generated projective R-module, then the localization P_p is free, that is, $P_p \cong (R_p)^n$ for some n.

Furthermore, a (finite) subcollection of the open sets $D(s)$ of $\mathrm{Spec}(R)$ may be chosen, say $\{D(s_i)\}$, to cover R such that $P_{s_i} \cong (R[\frac{1}{s_i}])^{n_i}$ for each i.

Proposition 1.6. All projective modules over an Euclidean domain are free.

1.2 Constructing Projective Modules

To construct more interesting examples of projective modules, we have the following "recipes" inspired from the contruction of a vector bundle in the topological case.

1.2.1 Open Patching

Let R be a commutative ring. Let $s_1, \dots s_n \in R$ be such that $s_1R + \cdots + s_nR = R$. That is, $D(s_i) \cong \operatorname{Spec}(R[\frac{1}{s_i}])$ cover $\operatorname{Spec}(R)$. Suppose we are given $g_{ij} \in GL_n(R[\frac{1}{s_is_j}])$ following the *cocycle conditions:*

1. $g_{ii} = 1$ for all i.
2. $g_{ij}g_{jk} = g_{ik}$ in $GL_n(R[\frac{1}{s_is_js_k}])$ for every i, j, k.

Then

$$P = \left\{ (x_1, \dots, x_c) \in \prod_{i=1}^{c} \left(R\left[\frac{1}{s_i}\right] \right)^n : g_{ij}(x_j) = x_i \text{ in } R\left[\frac{1}{s_is_j}\right]^n \text{ for all } i, j \right\}$$

is a finitely generated projective R-module.

1.3 Swan's Theorem

The following result due to Swan [Swa62] shows why projective modules are the correct algebraic analogue to vector bundles in the topological setting.

Theorem 1.7. *Let $\mathbb{F}$ be either $\mathbb{R}$ or $\mathbb{C}$. Let X be a compact Hausorff space, and write R for $C^0(X, \mathbb{F})$. Consider the* global sections functor *$\Gamma : E \mapsto \Gamma(E, \mathbb{F})$ from vector bundles over X to projective modules over R. Then Γ is an equivalence of categories.*

The algebraic analogue of vector bundles are *locally free sheaves over schemes*, which are an enrichment of the category of projective modules over unital rings.

Chapter 2

The Grothendieck Group K_0

In this chapter, we will develop the notion of the an abelian group K_0 called the Grothendieck Group accross a variety of contexts, which describes the additive properties of (right) projective modules over R.

We define the grothendieck group K_0 of a ring, of symmetric monoidal categories, and finally of exact cateogries. $K_0(R)$ corresponding to any unital ring R can be defined using all of these settings, and we shall show they are equivalent. We will also show the defined $K_0(R)$ group is functorial.

2.1 K_0 of a ring

Let R be a (unital) ring. The interesting objects of study will be finitely generated projective modules over R. We denote by $\mathbf{P}(R)$ the category of finitely generated projective modules over R. Note that the isomorphism classes of $\mathbf{P}(R)$ (if they form a set) derive a monoid structure through $\oplus$. This monoid is also abelian due to the isomorphisms $M \oplus N \cong N \oplus M$ for any R-modules M and N.

The idea is to consider the group naturally derived by implementing inverses for all elements in this monoid.

2.1.1 Group Completion of a Monoid

Given an (abelian) monoid $(M, +)$, we can derive an (abelian) group associated to M, which we shall call $\tilde{M}$, via the following procedure:, traingular Define an equivalence relation on $M \times M$ by

$$(a, b) \sim (c, d) \Leftrightarrow a + d + e = b + c + e \text{ for some } e \text{ in } M.$$

This indeed gives us an equivalence relation. Let $\widetilde{M}$ be the set of equivalence classes $[(m, n)]$ of $M \times M$ under this relation. This set admits a

natural (abelian) group operation defined by

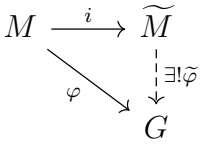
$$[(a,b)] + [(c,d)] := [(a+c, b+d)]$$

This is a well-defined, abelian and associative operation on $\widetilde{M}$ with the identity element $0 := [(0,0)]$ represented by any (l,l). Each element $[(a,b)]$ admits an inverse given by $[(b,a)]$ as $[(a,b)] + [(b,a)] = [(a+b, b+a)] = 0$.

A very familiar example of this construction is the abelian group $\mathbb{Z}$, which is the group completion of the abelian monoid $\mathbb{N}$ under $+$.

This construction is natural in M and follows a suitable universal property which we shall state.

Proposition 2.1 (Universal property of Group Completion)**.** We have a monoid homomorphism $i : M \to \widetilde{M}$ such that for any (abelian) group G, and a monoid homomorphism $\varphi : M \to G$, we have that φ must factor uniquely through i. That is, there is a unique map $\widetilde{\varphi} : \widetilde{M} \to G$ such that the following diagram commutes:



Proof. The monoid homomorphism $i : M \to \widetilde{M}$ is simply given by mapping a to $[(a,0)]$. This is indeed a homomorphism since $[(a,0)] + [(b,0)] = [(a+b,0)]$. Now suppose we are given $\varphi : M \to G$. Then $\widetilde{\varphi} : \widetilde{M} \to G$ is given by $[(a,b)] \mapsto \varphi(a) - \varphi(b)$, making the diagram commute. This map is unique since for any $a \in M$, we have $\widetilde{\varphi}([(a,0)]) = \varphi(a)$, and consequently $\widetilde{\varphi}([(0,a)]) = -\widetilde{\varphi}([(a,0)]) = -\varphi(a)$, which altogether determines the map $\widetilde{\varphi}$. □

Furthermore, any abelain group satisfying the given universal property is isomorphic to $\widetilde{M}$ via a unique isomorphism.

There is an alternative (but equivalent) characterization of $\widetilde{M}$ via the quotienting the free group on M by the monoidal relations.

Proposition 2.2. Let $K(M)$ be the abelian group given by taking the free abelian group $F(M)$ on the set M and quotienting by all relations of the form $[a] + [b] - [a+b]$ where $a, b \in M$ and $[x]$ is the symbol in $F(M)$ corresponding to the element $x \in M$. Then $K(M) \cong \widetilde{M}$ naturally.

Proof. We contruct a monoid homomorphism $j : M \to K(M)$ and use the univeresal property of $\widetilde{M}$ to conclude the result.

Define $j(m)$ as the image of the symbol $[m]$ in $K(M)$. Then note that We specify a group homomorphism $\Phi : K(M) \to \widetilde{M}$ by mapping the image of $[m]$ to $[(m, 0)]$. Note this is a well-defined homomorphism since $[(m, 0)] + [(n, 0)] = [(m + n, 0)]$. Furthermore, it is compatible with j, i.e. $\Phi \circ j = i$. Φ is clearly surjective. For injectivity, note that if $[(m, 0)] = [(0, 0)]$, then there is some $e \in M$ such that $m + e = e$. But this implies the image of m in $K(M)$ is $[m] = [m + e] - [e] = [e] - [e] = [0]$.

Due to the universal property of group completion, this isomorphism is natural.

□

Example 2.1.1. Let R be a (unital) ring satisfying the *Invariant Basis Property (IBP)*. That is, $R^m \cong R^n$ implies $m = n$. Then the monoid of isomorphism classes of free R-modules with the operation $\oplus$ is isomorphic to $\mathbb{N}$, the isomorphism given by the map $R^n \mapsto n$. Consequently, its group completion is isomorphic to the abelian group $\mathbb{Z}$.

Note that all fields (and division rings) F satisfy IBP since dimension of an F-module is independent of the choice of basis. Further, any ring R with a map $R \to F$, a field or a division ring F satisfies IBP. In particular, commutative rings all satisfy IBP.

Example 2.1.2. Let X be a topological space. Then the set $[X, \mathbb{N}]$ of continuous maps $X \to \mathbb{N}$ forms an abelian monoid under pointwise addition of maps. Then the group completion of this monoid is isomorphic to the abelian group of continuous maps $X \to \mathbb{Z}$ denoted $[X, \mathbb{Z}]$. If X is quasicompact, this is in fact a free abelian group.

2.1.2 $K_0(R)$ of a ring R

Definition 2.3 ($K_0(R)$)**.** Now we define the *Grothendieck group* $K_0(R)$ for a unital ring R as the group completion of the monoid $P(R)$ of isomorphism classes of finitely generated projective R-modules with the monoid operation given by $\oplus$.

Example 2.1.3. Let F be a field or a division ring. Then all finitely generated projective F-modules are free. This gives us $K_0(F) \cong \mathbb{Z}$. More generally, for any local ring R, all finitely generated projective R-modules are free. This gives us $K_0(R) \cong \mathbb{Z}$.

Example 2.1.4. Note that finitely generated projectives over Euclidean Domains, in particular over $\mathbb{Z}$ are free. Thus $K_0(\mathbb{Z}) \cong \mathbb{Z}$. More generally, for any ring R with the IBP, we have an injective group homomorphism $\mathbb{Z} \to K_0(R)$ given by $n \mapsto R^n$.

Here is an observation which shows why we restrict our attention to finitely generated projective modules.

Example 2.1.5 (Eilenberg Swindle)**.** Let R^∞ be an infinitely generated free module over R. Suppose P is a projective module over commutative ring R, such that $P \oplus Q \cong R^\infty$. In fact, such a R^∞ and Q exists for any projective P. Then

$$\begin{aligned} P \oplus R^\infty &\cong P \oplus (Q \oplus P) \oplus (Q \oplus P) \oplus \cdots \\ &\cong R^\infty \end{aligned}$$

If we had taken considered the group completion of the category of all projective modules over R, we would have $[P] = 0$ in K_0, and $K_0(R) = 0$.

2.1.3 Morita Equivalence and Invariance

Suppose we have (unital) rings R and S such that we have an equivalence of the respective categories of right modules:

$$\textbf{mod-}R \simeq \textbf{mod-}S$$

Then we say that R and S are **Morita equivalent**.

A consequence of Morita Equivalence is that R and S yield isomorphic K_0 groups. An important example of Morita Equivalence is

Proposition 2.4 (Morita Invariance of K_0)**.** For any unital ring R, we have

$$\textbf{mod-}R \simeq \textbf{mod-}M_n(R)$$

and consequently

$$K_0(R) \cong K_0(M_n(R))$$

2.1.4 Base Change and Transfer

Suppose we have a ring homomorphism $f : R \to S$ of unital rings. Then this induces a *base change functor* $f^* : \mathbf{P}(R) \to \mathbf{P}(S)$ by extension of scalars $P \mapsto P \otimes_R S$. This functor is additive in the sense that

$$f^*(P \oplus Q) \cong f^*(P) \oplus f^*(Q)$$

In particular, if $P \cong Q$, then $f^*(P) \cong f^*(Q)$. This gives us a group homomorphism

$$K(f^*) : K_0(R) \to K_0(S)$$

Note that this implies that K_0 is a (covariant) functor from **Ring** $\to$ **Ab**.

Conversely, we also have a *transfer functor* $f_* : \mathbf{P}(S) \to \mathbf{P}(R)$ given by restriction of scalars, which is also additive and gives us $K(f_*) : K_0(S) \to K_0(R)$.

2.2 K_0 of a Symmetric Monoidal Category

We will now introduce the K_0 group in a different setting, called *symmetric monoidal categories*, which is an abstraction of some of the common properties that $\oplus, \otimes, \sqcup, \times$ share.

Definition 2.5. A *symmetric monoidal category* is a category S with a distinguished object e and a functor $\Box : S \times S \to S$ satisfying natural isomorphisms

$$e\Box s \cong s, \qquad s\Box e \cong s \qquad s\Box(t\Box u) \cong (s\Box t)\Box u, \quad \text{and} \quad c_{s,t} : s\Box t \cong t\Box s$$

with $c_{s,t} \circ c_{t,s} = 1_{t\Box s}$ for all $s, t, u \in S$. Furthermore, these natural isomorphisms must obey coherence laws given by the MacLane pentagon for associativity:

$$\begin{array}{ccccc} a\Box(b\Box(c\Box d)) & \xrightarrow{\cong} & (a\Box b)\Box(c\Box d) & \xrightarrow{\cong} & ((a\Box b)\Box c)\Box d \\ \downarrow{\scriptstyle\cong} & & & & \uparrow{\scriptstyle\cong} \\ a\Box((b\Box c)\Box d) & & \xrightarrow{\cong} & & (a\Box(b\Box c)\Box d), \end{array}$$

as well as the unit triagular laws

$$\begin{array}{ccc} a\Box(e\Box c) & \xrightarrow{\cong} & (a\Box e)\Box c \\ & {\scriptstyle\cong}\searrow \quad \swarrow{\scriptstyle\cong} & \\ & a\Box c & \end{array}$$

and the commutativity hexagon

$$\begin{array}{ccccc} a\Box(b\Box c) & \xrightarrow{\cong} & (a\Box b)\Box c & \xrightarrow{\cong} & c\Box(a\Box b) \\ \downarrow{\scriptstyle\cong} & & & & \downarrow{\scriptstyle\cong} \\ (a\Box c)\Box b & \xrightarrow{\cong} & (a\Box c)\Box b & \xrightarrow{\cong} & (c\Box a)\Box b \end{array}$$

for objects a, b, c, d in S and appropriate arrows in each diagram.

The role of coherence is so that we can write products $s_1\Box s_2\Box s_3\Box\cdots\Box s_n$ unambiguously and consistently. More precisely, there is a consistent and natural way to transform a given association and permutation of the product into any other.

Through the rest of this text, we will mostly gloss over concerns about coherence when dealing with symmetric monoidal categories. The reader is suggested to refer to [Mac63] for the precise description of symmetric monoidal categories and why the pentagonal, traingular and hexagonal diagrams suffice for coherence.

Example 2.2.1. Any category with either a direct sum $\oplus$ is symmetric monoidal with $\square = \oplus$, which includes all abelian categories. In particular, for any unital ring R,

- $\mathbf{F}(R)$ (category of finitely generated free modules over R),
- $\mathbf{P}(R)$ (category of finitely generated projective modules over R),

are both symmetric monoidal with $\square = \oplus$. the category of all finitely generated modules over R are all examples of symmetric monoidal categories.

Categories with finite coproducts $\sqcup$ as well as categories with finite products similarly are symmetric monoidal with $\square = \sqcup$ and $\square = \times$ respectively.

Example 2.2.2. Given a topological space X, the category of (real or complex) vector bundles is symmetric monoidal with $\square = \oplus$.

Definition 2.6 (K_0 of a symmetric monoidal category)**.** Suppose that isomorphism classes of a symmetric monoidal category S form a set. Then we can define the *Grothendieck group* $K_0^{\square}(S)$ to be the abelian group generated by isomorphism classes of S, quotiented by relations given by $[s \square t] = [s]+[t]$ for every object s, t in S.

When it is understood, we may drop $\square$ and write $K_0(S)$.

Proposition 2.7. $K_0(S)$ is the group completion of the (abelian) monoid of isomorphism classes of S.

Example 2.2.3. Given a commutative ring R, the category $\mathbf{Pic}(R)$ of algebraic line bundles (invertible modules) over R is symmetric monoidal with $\square = \otimes$ and $e = R$. Since $\mathbf{Pic}(R)$ is already a group, we have $K_0(\mathbf{Pic}(R)) = \mathbf{Pic}(R)$.

2.3 K_0 of an Exact Category

We now describe K_0 in yet another context. The notion of an exact category was formalized by Quillen in [Qui73], where the notion is helpful for building higher K-Theory which we shall see in Chapter 5.

Definition 2.8 (Exact category)**.** An exact category is a pair $(\mathcal{C}, \mathcal{E})$ where $\mathcal{C}$ is an additive category fully embedded in some ambient abelian category $\mathcal{A}$, and $\mathcal{E}$ is the class of all sequences

$$0 \to M' \to M \to M'' \to 0 \tag{2.1}$$

in $\mathcal{C}$ which are exact in $\mathcal{A}$, such that $\mathcal{C}$ is closed under extensions, i.e., for every exact sequence (2.1) in $\mathcal{A}$, we have that M is in $\mathcal{C}$ if and only if M' and M'' are both in $\mathcal{C}$.

From here on, we will simply write $\mathcal{C}$ for an exact category where $\mathcal{E}$ and $\mathcal{A}$ are implicit.

Definition 2.9 (Admissible mono and epimorphisms)**.** If a morphism $C' \to C$ appears in $\mathcal{E}$, we say it is an *admissible monomorphism*. Dually, if $C \to C''$ appears in $\mathcal{E}$, we say it is an *admissible epimorphism*.

Admissible monomorphisms and epimorphisms will be denoted $C' \rightarrowtail C$ and $C \twoheadrightarrow C''$ respectively.

Proposition 2.10. We note the following properties about the class $\mathcal{E}$ of exact sequences.

1. Any sequence isomorphic to a sequence in $\mathcal{E}$ is in $\mathcal{E}$. An isomorphism of exact sequences is a commutative diagram
2. For any C and D, the sequence

$$0 \to C \xrightarrow{(\mathrm{id},0)} C \oplus D \xrightarrow{pr_2} D \to 0$$

   is in $\mathcal{E}$.
3. The class of admissible epimorphisms (resp. monomorphisms) is closed under composition.
4. Let $C \to C''$ be a map possessing a kernel in $\mathcal{C}$. If there exists a map $C'' \to C$ such that $C'' \to C \to C$ is an admissible epimorphism, then $C \to C''$ is an admissible epimorphism. Dually, for monomorphisms.
5. Suppose we have a sequence $0 \to C' \to C \to C''$ in $\mathcal{E}$ and a map $f : D \to C''$ in $\mathcal{C}$. The pullback $E = C \times_{C''} D$ leads to the diagram

$$\begin{array}{ccccccccc} 0 & \longrightarrow & C' & \longrightarrow & E & \xrightarrow{pr_2} & D & \longrightarrow & 0 \\ & & \| & & \downarrow{\scriptstyle pr_1} & & \downarrow{\scriptstyle f} & & \\ 0 & \longrightarrow & C' & \longrightarrow & C & \longrightarrow & C'' & \longrightarrow & 0 \end{array}$$

   and $E \to C$ is admissible.

All these properties are immediate from the fact that $\mathcal{C}$ is closed under extensions and $\mathcal{E}$ is maximal.

Definition 2.11 (Exact functors)**.** Let $\mathcal{C}$ and $\mathcal{B}$ be exact categories. A functor $F : \mathcal{B} \to \mathcal{C}$ is called *exact* if it takes exact sequences in $\mathcal{B}$ to exact sequences in $\mathcal{C}$.

Example 2.3.1. An abelian category is exact in the evident way.

We may now define the Grothendieck Group on an exact category $\mathcal{C}$ by requiring additivity relations on exact sequences.

Definition 2.12 (K_0 of an exact category)**.** Let $\mathcal{C}$ be a small exact category. Then we define $K_0(\mathcal{C})$ as the abelian group generated by symbols $[C]$, one for each object C in $\mathcal{C}$, and relations $[C] = [C'] + [C'']$ for every exact sequence

$$0 \to C' \to C \to C'' \to 0$$

It follows that $[0] = 0$, $[C \oplus D] = [C] + [D]$ and for any $C \cong D$, we have $[C] = [D]$.

Example 2.3.2. The category $\mathbf{P}(R)$ is an exact category via its embedding in R-**mod**. Since every exact sequence of projective modules splits, we have $K_0(\mathbf{P}(R)) = K_0(R)$.

Definition 2.13 (Split-exact categories)**.** An exact category $\mathcal{C}$ is said to be *split-exact* if every short exact sequence in $\mathcal{E}$ is split, i.e., isomorphic to a sequence $0 \to C \to C \oplus D \to D \to 0$. In this case, this $K_0(\mathcal{C}) \cong K_0^{\oplus}(\mathcal{C})$ naturally if $\mathcal{C}$ is viewed as symmetric monoidal with respect to $\oplus$.

$\mathbf{P}(R)$ is an example of a split-exact category. It will turn out in Chapter 5 that for such categories, the higher K_n-groups defined are also isomorphic.

Chapter 3

Groups K_1 and K_2 of a ring

We will now define the classical theory of $K_1(R)$ and $K_2(R)$ groups for a unital ring R. As with $K_0(R)$, their definitions require only elementary linear algebra and group theory.

3.1 The Group K_1 of a ring

Let R be a unital ring. Consider the groups $GL_n(R)$ for $n \geq 0$. One may inject each $GL_n(R)$ into $GL_{n+1}(R)$ via the embedding

$$\begin{aligned} GL_n(R) &\hookrightarrow GL_{n+1}(R) \\ g &\mapsto \begin{pmatrix} g & 0 \\ 0 & 1 \end{pmatrix} \end{aligned}$$

This gives us a filtered system of groups

$$R^\times \cong GL_1(R) \subset GL_2(R) \subset \cdots \subset GL_n(R) \subset GL_{n+1}(R) \subset \cdots$$

Definition 3.1. Taking the colimit of this filtered system gives us the *stable general linear group*, which we denote $GL(R)$.

Definition 3.2 (Elementary matrices and subgroup)**.** Suppose $i \neq j$, the the elementary matrix e_{ij} is the matrix in $GL_n(R)$ given by $e_{ij}(r) = I_n + rE_{ij}$, that is, identity along the diagonal and r in the (i, j)-th spot.

The subgroup of $GL_n(R)$ generated by $e_{ij}(r)$ is called the *elementary subgroup*, denoted $E_n(R)$.

Similar to $GL(R)$, we have the embeddings $E_n(R) \hookrightarrow E_{n+1}(R)$ given by

$$\begin{aligned} E_n(R) &\hookrightarrow E_{n+1}(R) \\ e &\mapsto \begin{pmatrix} e & 0 \\ 0 & 1 \end{pmatrix} \end{aligned}$$

This gives us another filtered system of groups

$$1 \cong E_1(R) \subset E_2(R) \subset \cdots \subset E_n(R) \subset E_{n+1}(R) \subset \cdots$$

Definition 3.3. The colimit of this filtered system gives us the *stable elementary subgroup*, denoted $E(R)$, which may be identified with a subgroup of the stable group $GL(R)$.

We prove an elementary property about elementary matrices.

Lemma 3.4. Suppose g is an element in $GL_n(R)$. Then $\left(\begin{smallmatrix} g & 0 \\ 0 & g^{-1} \end{smallmatrix}\right)$ is in $E_{2n}(R)$.

Proof. Note that matrices of the form $\left(\begin{smallmatrix} 1 & g \\ 0 & 1 \end{smallmatrix}\right)$ are in $E_{2n}(R)$. Consequently, we have the factorization

$$\begin{pmatrix} g & 0 \\ 0 & g^{-1} \end{pmatrix} = \begin{pmatrix} 1 & g \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ g^{-1} & 1 \end{pmatrix} \begin{pmatrix} 1 & g \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$$

so the given element is in $E_{2n}(R)$. □

Lemma 3.5. $E(R)$ is a normal subgroup of $GL(R)$.

Proof. We have for any $e \in E_n(R)$ and $g \in GL_n(R)$,

$$\begin{pmatrix} geg^{-1} & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} g & 0 \\ 0 & g^{-1} \end{pmatrix} \begin{pmatrix} e & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} g^{-1} & 0 \\ 0 & g \end{pmatrix}$$

Since each of the elements are in $E_{2n}(R)$, the product is in $E_{2n}(R)$. This implies $E(R)$ is normal in $GL(R)$. □

Now we arrive at our definition of $K_1(R)$.

Definition 3.6 ($K_1(R)$)**.** We define group $K_1(R)$ to be the quotient

$$K_1(R) := GL(R)/E(R)$$

Example 3.1.1. If F is a field, then $K_1(F) = F^\times$.

3.1.1 Alternative Definition of K_1

Proposition 3.7. For elementary matrices, we have $e_{ij}(r)e_{ij}(s) = e_{ij}(r + s)$, and $e_{ij}(r)$ has the inverse $e_{ij}(-r)$. Otherwise, we have the following commutation relations between arbitrary elementary matrices:

$$[e_{ij}(r), e_{kl}(s)] = \begin{cases} 1 & \text{if } j \neq k \text{ and } i \neq l \\ e_{il}(rs) & \text{if } j = k \text{ and } i \neq l \\ e_{kj}(-sr) & \text{if } j \neq k \text{ and } i = l \end{cases}$$

We say that a group G is *perfect* if $G = [G, G]$. Proposition 3.7 immediately gives us that

Lemma 3.8. If $n \geq 3$, then $E_n(R)$ is perfect.

Proof. If i, j, k are distinct, we have $e_{ij}(r) = [e_{ik}(r), e_{kj}(1)]$ which shows $E_n(R) \subseteq [E_n(R), E_n(R)]$, giving us the result. □

This also shows that the commutator subgroup $[GL(R), GL(R)]$ contains all elements of $E(R)$ after stabilizing, since every element in $E(R)$ can be thought of as an element in $E_n(R)$ for some large n. Here is a result due to Whitehead's 1950 paper [Whi50]

Lemma 3.9 (Whitehead's Lemma)**.** We have $E(R) = [GL(R), GL(R)]$. In particular, $E(R)$ is perfect, and $K_1 = GL(R)_{ab}$.

Proof. Since the commutator subgroup contains $E(R)$, it suffices to show the converse. Every commutator in $GL_n(R)$ may be written as the following product

$$[g, h] = \begin{pmatrix} g & 0 \\ 0 & g^{-1} \end{pmatrix} \begin{pmatrix} h & 0 \\ 0 & h^{-1} \end{pmatrix} \begin{pmatrix} (hg)^{-1} & 0 \\ 0 & hg \end{pmatrix}$$

But each of the elements in the product are in $E_{2n}(R)$. This gives us the desired claim. □

Clearly, this also means that $K_1(R)$ is abelian for any ring R. Note that in some texts, $K_1(R) = GL(R)_{ab}$ is taken to be the definition. and Whitehead's Lemma implies the converse.

3.1.2 Homological Interpretaion of K_1

K_1 may be interepreted in the language of homological algebra. We refer the reader to [Wei94] Note that the first group homology $H_1(G; \mathbb{Z})$ of any group G is naturally isomorphic to $G/[G, G]$. This gives us

$$K_1(R) = H_1(GL(R); \mathbb{Z}) \tag{3.1}$$

This relates to the fact the the group $K_2(R)$ which we will define later can also be interpreted in terms of Homological Algebra.

We also state Bass's definition of K_1 in [Bas68].

Definition 3.10. Define the *classical translation category* $t\mathbf{P}(R)$ of $\mathbf{P}(R)$ as follows. The objects of $t\mathbf{P}(R)$ are isomorphism classes of finitely generated projective modules, and the morphisms $P \to P'$ are isomorphisms classess of Q such that $P \oplus Q \cong P'$.

Note that this is a filtered category, and $P \mapsto H_1(\mathrm{Aut}(P); \mathbb{Z})$ is functorial.

Definition 3.11. We have $K_1^{\mathrm{Bass}}(R) := \varinjlim_{P \in t\mathbf{P}} H_1(\mathrm{Aut}(P); \mathbb{Z})$

Theorem 3.12. *For any unital ring R, we have $K_1(R) \cong K_1^{Bass}(R)$.*

Consequently, we have this alternative presentation for K_1:

Theorem 3.13. *$K_1(R)$ may also be described as the group generated by symbols $[(P, \alpha)]$ where P is a (right) projective module over R and $\alpha : P \xrightarrow{\cong} P$ is an automorphism, subject to the following relations:*

1. *$[(P, \alpha)] = [(Q, \beta)]$ if there is an $f : P \xrightarrow{\cong} Q$, for which $f\alpha = \beta f$.*
2. *$[(P, \alpha \circ \beta)] = [(P, \alpha)][(P, \beta)]$*
3. *$[(P \oplus Q, \alpha \oplus \beta)] = [(P, \alpha)][(Q, \beta)]$*

The proof follows almost immediately from the definition of K_1^{Bass}.

Note that the relations force $K_1(R)$ to be abelian.

Remark 3.1.1. The group K_1 was studied by Whitehead in the theory of simple homotopy types and whitehead torsion. The Whitehead group, which arises naturally in this context, is a quotient of $K_1(\mathbb{Z}[\pi])$. The Whitehead group $Wh_1(\pi)$ of a (fundamental) group π is the quotient $K_1(\mathbb{Z}[\pi])/H$ where H is the subgroup of $K_1(\mathbb{Z}[\pi])$ generated by $+-1$ and $GL_1(\pi)$.

3.2 K_2 of a ring

The notion of the K_2 group of a ring was introduced by Milnor in 1967. We refer the reader to [Mil71] for a detailed discussion of the subject. Here, we shall define K_2 and state important results.

We showed that the group $E_n(R)$ obeyed certain relations in Proposition 3.7. We define a group which models these relations.

Definition 3.14 ($K_2(R)$)**.** Let $n \geq 3$. The *Steinberg group* $St_n(R)$ of a ring R is defined as the group generated by the symbols $x_{ij}(r)$ for $1 \leq i \neq j \leq n$ and $r \in R$ subject to the relations

$$x_{ij}(r)x_{ij}(s) = x_{ij}(r+s) \tag{3.2}$$

$$[x_{ij}(r), x_{kl}(s)] = \begin{cases} 1 & \text{if } j \neq k \text{ and } i \neq l \\ x_{il}(rs) & \text{if } j = k \text{ and } i \neq l \\ x_{kj}(-sr) & \text{if } j \neq k \text{ and } i = l \end{cases} \tag{3.3}$$

Note that these relations are satisfied by elementary matrices. This gives us a surjective homomorphism $St_n(R) \to E_n(R)$.

It is clear we have another filtered system given by embeddings $St_n(R) \hookrightarrow St_{n+1}(R)$, the colimit of which we call the *stable Steinberg group*, denoted $St(R)$.

Since we can think of $E_n(R)$ as living in $E(R)$, we have homomorphisms $St_n(R) \to E_n(R) \to E(R)$, which induce a (surjective) group homomorphism $\pi : St(R) \to E(R)$.

Definition 3.15 (K_2 of a ring)**.** We define the group $K_2(R)$ for a unital ring R as the kernel of $\pi : St(R) \to E(R)$.

Note that this gives us an exact sequence

$$1 \to K_2(R) \to St(R) \xrightarrow{\pi} E(R) \to 1 \tag{3.4}$$

which completes into a longer sequence

$$1 \to K_2(R) \to St(R) \to GL(R) \to K_1(R) \to 1 \tag{3.5}$$

We state a result due to Steinberg. (See [Ste67])

Theorem 3.16 (Steinberg)**.** *$K_2(R)$ is precisely the center of $St(R)$, hence abelian.*

3.2.1 Universal Central Extensions

The Steinberg group $St(R)$ can be described in terms of universal central extensions, which we shall define. See [Mil71] and [Suz82]. From the viewpoint of homological algebra, central extensions of G are classified by the cohomology group $H^2(G;\mathbb{Z})$ [Wei94], and it turns out one can describe $K_2(R)$ as a homology group, $K_2(R) \cong H_2(E(R);\mathbb{Z})$. Here, we give a skeleton of the results leading up to this fact.

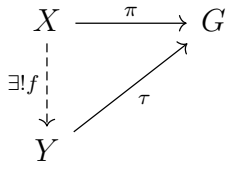
Definition 3.17. A *central extension* of a group G is an exact sequence

$$1 \to A \to X \to G \to 1$$

such that the image of A is in the center of H.

We will often simply write $X \xrightarrow{\pi} G$ to denote the central extension when A is clear.

Definition 3.18. A *univeresal central extension* of G is a central extension which is universal among central extensions of G. That is, for any central extension $Y \xrightarrow{\tau} G$ we have a unique group homomorphism $f : X \to Y$



Clearly, a universal central extension is unique up to a unique isomorphism if it exists.

Lemma 3.19. If G has a universal central extension $X \xrightarrow{\pi} G$, then G and X are both perfect.

Proof. If not, then $B = X/[X, X]$ is nontrivial, and we have the central extension $B \times G \to G$ given by the projection and two homomorphisms $X \to B \times G$, given by $(0, \pi)$ and (pr, π) compatible with the extension, where $pr : X \to [X, X]$ is the natural projection. □

We also have a converse.

Let a group G have a presentation F/R where F is a free group and R is the subgroup of relations. This may be described as an exact sequence

$$1 \to R \to F \to G \to 1$$

Since $[R, F]$ is a normal subgroup of F, we get the following central extensions.

$$1 \to R/[R, F] \to F/[R, F] \to G \to 1$$

$$1 \to (R \cap [F, F])/[R, F] \to [F, F]/[R, F] \to [G, G] \to 1 \tag{3.6}$$

Note the group $(R \cap [F, F])/[R, F]$ is exactly the homology group $H_2(G; \mathbb{Z})$.

Theorem 3.20 (Recognition Theorem for Universal Central Extensions)**.** *Every perfect group G has a universal central extension*

$$1 \to H_2(G; \mathbb{Z}) \to [F, F]/[R, F] \to G \to 1$$

Consequently, if X is any central extension of G, then the following are equivalent

1. *X is a universal central extension.*
2. *X is perfect and every central extension of X splits.*
3. *$H_1(X; \mathbb{Z}) = H_2(X; \mathbb{Z}) = 0$*

See [Mil71] for a complete proof. The following result follows. (See [Ker70])

Theorem 3.21 (Kervaire, Steinberg)**.** *The Steinberg group $St(R)$ is the universal central extension of $E(R)$. Thus we can conclude*

$$K_2(R) \cong H_2(E(R); \mathbb{Z})$$

3.2.2 Bass's K_2

We also state the definition of $K_2(R)$ due to Bass, which turns out to be equivalent to our definition of $K_2(R)$. (See [Bas68], [**Kervaire-Steinberg**])

Definition 3.22 (Bass's K_2)**.** $K_2^{\text{Bass}}(R) := \varinjlim_{P \in t\mathbf{P}} H_2([\text{Aut}(P), \text{Aut}(P)]; \mathbb{Z})$

Theorem 3.23. $K_2(R) \cong K_2^{Bass}(R)$

In Chapter 5, we shall discover homotopy-theoretic constructions which yield the groups $K_n(R)$ and prove their equivalence with these classical notions. However, it will turn out that these higher K_n groups defined by Quillen are not isomorphic to the K_n groups defined by Bass using homological algebra.

Chapter 4

Homotopy Theory Preliminaries

In this chapter, we shall explore various ideas from Homotopy Theory that shall be relevant for construction higher K-theory in the next chapter.

In Section §1, we first recall some basic notions, and introduce homotopy fiber seqeunces. In Section §2, we describe a homotopy theoretic properties of small categories, and prove Quillen's Theorems A and B from [**Quillen**]. In Section §3, we introduce and mention some properties of H-spaces. In Section §4, we describe the +-construction on CW complexes. We mention the characterization of the +-construction by acyclic maps and state the Theorem due to Quillen proving uniqueness up to homotopy of the +-construction. We finally also give an explicit +-construction by attaching 2-cells and 3-cells.

4.1 Basic Notions

A based space is a topological space X along with a distinguished basepoint which we denote $*_X$. A map of based spaces $X \to Y$ is a map of the underlying topological spaces which preserves basepoints.

We can functorially associate homotopy groups π_n to a based space X, denoted $\pi_n(X)$. We assume familiarity with the definitions of homotopy and homology groups. $H_*(X; M)$ will refer to the singular homology of X with coefficients in some abelian group M, or alternatively local coefficients, i.e. M being a $\pi_1(X)$-module. When $M = \mathbb{Z}$, this is referred to as integral homology.

A CW complex (or cellular complex) X is a colimit of a filtration of n-skeleta, denoted $\emptyset = X_{-1} \subseteq X_0 \subseteq X_1 \subseteq X_2 \subseteq \cdots$ where X_0 is a discrete

set of points and X_n is a pushout of X_{n-1} via a disjoint union $\cup_\alpha \varphi_\alpha$ of attaching maps $\varphi_\alpha : S^{n-1}_\alpha \to X_{n-1}$ and the disjoint union $\cup_\alpha j_\alpha$ of inclusions $j_\alpha : S^{n-1}_\alpha \to D^n_\alpha$. (Note S^{-1} is taken to be $\emptyset$.)

That is, $X_n = X_{n-1} \cup_{S^{n-1}_\alpha} D^n_\alpha$. The images of the interiors of the disks D^n_α in X_n are called n-cells. Note that 0-cells correspond to a set of points, 1-cells correspond to line segments, 2-cells to disks, and so on.

The colimit carries the weak topology. The homology of CW complexes is particularly tractable. See [Hat02] for a detailed treatment.

4.1.1 Homotopy Fiber and Fiber Sequences

We now describe the important notion of a *homotopy fiber*, which is useful for organizing how homotopy groups behave across a map $f : E \to B$.

Definition 4.1 (Homotopy Fiber)**.** Let $f : E \to B$ be a map of based spaces. The homotopy fiber $F(f)$ of this map is the homotopy-pullback of the cospan $E \to B \leftarrow PB$, where PB is the (based) space of paths in Y.

More concretely, we can define $F(f)$ to be consisting of pairs (e, γ) where $e \in E$ and γ is a path in B starting at the basepoint such that $f(e) = \gamma(1)$. $F(f)$ inherits the subspace topology from $E \times PB$ where PB itself carries the compact-open topology. We may declare $F(f)$ to have the basepoint $(*_E, *_{PB})$.

The key property of the homotopy fiber is that the sequence $F(f) \to E \to B$ of based space maps which yields a long exact sequence:

$$\begin{aligned} \cdots \to \Omega^{n+1}B \xrightarrow{\partial} \Omega^n F(f) \to \Omega^n E \to \Omega^n B \xrightarrow{\partial} \cdots \\ \cdots \xrightarrow{\partial} \Omega F(f) \to \Omega E \to \Omega B \xrightarrow{\partial} F(f) \to E \to B \end{aligned}$$

which in turn yields a long exact sequence of groups/pointed sets:

$$\begin{aligned} \cdots \to \pi_{n+1}B \xrightarrow{\partial} \pi_n F(f) \to \pi_n E \to \pi_n B \xrightarrow{\partial} \cdots \\ \cdots \xrightarrow{\partial} \pi_1 F(f) \to \pi_1 E \to \pi_1 B \xrightarrow{\partial} \pi_0 F(f) \to \pi_0 E \to \pi_0 B \end{aligned}$$

Any other such sequence $F \to E \to B$ which results in the evident map $F \to F(f)$ (use $*_{PB}$ in the image) being a homotopy equivalence, is called a *homotopy fibration sequence*, and consequently also yields such a long exact sequence.

If $f : E \to B$ were a homotopy equivalence, then the homotopy fiber $F(f)$ is weakly contractible, i.e., it has the homotopy groups of a point, $\pi_* F(f) \cong \pi_*(\star)$. If E and B have homotopy type of CW complexes, then so

is $F(f)$ and by Whitehead's Theorem [Whi78], weak homotopy equivalence implies homotopy equivalence.

Conversely, if the fiber $F(f)$ is contractible, then f is a weak homotopy equivalence, i.e., the induced maps $\pi_n(E) \to \pi_n(B)$ are isomorphisms. If E and B have homotopy type of CW complexes, then f is a homotopy equivalence.

4.1.2 Geometric Realization of a small category

Given a small category C, there is a way to interpret C as a CW complex by the following procedure:

Definition 4.2 (Simplicial sets)**.** A simplicial set X is a contravariant functor $\mathbf{\Delta} \to \mathbf{Sets}$, where $\mathbf{\Delta}$ is the simplex category consisting of objects $\mathbf{n} = \{0, 1, \ldots, n-1\}$ and morphisms $\mathbf{i} \to \mathbf{j}$ given by weakly-order preserving maps. Or alternatively, a sequence of sets $X_0, X_1, X_2, \ldots$ with face maps $\partial_i^n : X_n \to X_{n-1}$ and degeneracy maps $\sigma_j^n : X_n \to X_{n+1}$ subject to certain identities $(0 \leq i, j, \leq n)$.

The category of simplicial sets is denoted **sSet**. It is clear **sSet** is the functor category $[\mathbf{\Delta}^{op}, \mathbf{Sets}]$.

Example 4.1.1. The category Δ[1] of standard n-simplices Δ^n with morphisms being the standard inclusion maps $\Delta^i \to \Delta^j$ for $0 \leq i \leq j$ gives a simplicial set by $\mathbf{n} \to \Delta^n$ and $f : \mathbf{i} \to \mathbf{j}$ associated to $\Delta^i \to \Delta^j$. In fact, this is an isomorphism of categories $\Phi : \mathbf{\Delta} \cong \Delta$.

Example 4.1.2. Any space X determines a simplicial set by the representable functor $\mathbf{Top}(\Phi(-), X)$. This construction is clearly a (covariant) functor $S : \mathbf{Top} \to \mathbf{sSet}$. It is called the *singular functor*.

A simplicial set X may be functorially turned into a topological space, and this functor is called Geometric Realization.

Proposition 4.3 (Geometric Realization)**.** There exists a functor $|-| : \mathbf{sSet} \to \mathbf{Top}$ which is left-adjoint to the singular functor.

$$\mathbf{Top}(|X|, Y) \cong \mathbf{sSet}(X, SY)$$

naturally in X and Y.

We can define $|-|$ to be the functor taking X to the coend $\int_{\mathbf{n} \in \mathbf{\Delta}} X_n \times \Delta^n$, which is a disjoint union of $X_n \times \Delta^n$ for every n, with faces identified via appropraite face maps $\partial_i^n : \Delta^n \to \Delta^{n-1}$. See [Maclane]. Furthermore, $|X|$ has a natural CW structure induced by CW structures on Δ^n.

[1] not identical to $\mathbf{\Delta}$

Definition 4.4 (Nerve of a small category)**.** Let C be a small category. Then its nerve NC is a simplicial set defined by taking the n-simplices as functors $c : \mathbf{n} \to C$, i.e. diagrams of the form

$$c_0 \to c_1 \to \cdots \to c_{n-1}$$

for objects $c_0, c_1, \ldots, c_{n-1}$ in C. The i-th face map $\partial_i^n(c)$ on this simplex is obtained by deleting c_i in the evident way. Similarly, the j-th degeneracy map $\sigma_j^n(c)$ is obtained by replacing c_j by $c_j \to c_j$ in the obvious manner.

We define the *classifying space* of a small category C, denoted BC, as the geometric realization of NC.

Definition 4.5 (Classifying space of a small category)**.** $BC := |NC|$.

Example 4.1.3. For the categories $\mathbf{n}$, the geometric realization $B\mathbf{n} = \Phi\mathbf{n}$.

Furthermore, for any small category C, BC is the colimit (with the weak topology) of the directed system of the functors $B\mathbf{n} \to BC$ coming from $\mathbf{n} \to C$ and the morphisms coming from factorizations $\mathbf{i} \to \mathbf{n} \to C$.

Proposition 4.6. A functor $F : C \to D$ of small categories determines a cellular map $BF : BC \to BD$. Further, $B(\mathrm{id}_C) : BC \to BC$ is the identity map on BC, and for functors $F : C \to D, G : D \to E$, we have $B(G \circ F) = BG \circ BF$.

Proof. Due to the functoriality of geometric realization on simplicial sets, it suffices to show that the association $N : \mathbf{Cat} \to \mathbf{sSet}$ is functorial. But this follows directly from axioms of a category. □

Thus we have a functor $B : \mathbf{Cat} \to \mathbf{Top}$.

Example 4.1.4. A group G may be interpreted as a small category with a single object and all invertible arrows. We call BG the classifying space of the group G.

Proposition 4.7. The fundamental group $\pi_1(BG)$ is naturally isomorphic to G.

4.2 Homotopy Theoretic Aspects of Categories

Owing to the geometric realization functor $B : \mathbf{Cat} \to \mathbf{Top}$, we can think of small categories as topological spaces via B. This allows us to extend

topological notions to categories by viewing them through their geometric realization.

By abuse of notation, we will make statements such a category C is contractible, or connected, etc. (any homotopy-theoretic property) if its geometric realization is so. If a functor $F : C \to D$ induces a homotopy equivalence $BF : BC \to CD$, then we say that $F : C \to D$ is a homotopy equivalence, and the categories C and D are homotopy equivalent.

Proposition 4.8. A natural transformation $\eta : F \Rightarrow G$ between functors $F, G : C \to D$ describes a homotopy $BC \times [0,1] \to BD$ which we denote $B\eta : BF \Rightarrow BG$.

Proof. We can view $\eta : F \Rightarrow G$ as a functor $C \times \mathbf{2} \to D$ whose restriction to $C \times 0$ is F and the restriction to $C \times 1$ is G. Upon taking the geometric realization, we are done. □

A consequence is that an adjoint pairs of functors $L : C \to D, R : D \to C$ with natural transformations $LR \Rightarrow \mathrm{id}_D$ and $\mathrm{id}_C \Rightarrow RL$, leads to a homotopy equivalence $BC \simeq BD$. In particular,

Corollary 4.9. An equivalence of small categories is a homotopy equivalence.

Proposition 4.10. If a small category C has an initial or a final object, then BC is contractible.

Proof. Let C have an initial object c_0, then the (unique) functor $F : C \to \mathbf{1}$ has a left adjoint $L : \mathbf{1} \to C$ given by $0 \mapsto c_0$. Dually, if C has a final object c_1, then F has a right adjoint $R : \mathbf{1} \to C$ given by $0 \mapsto c_1$. The result follows from the remark made earlier. □

4.2.1 Quillen's Theorems A and B

In this section, we prove two theorems originally proven by Quillen in [Qui73], which shall be integral towards proving the $+ = Q$ Theorem in Chapter 5.

Theorem A gives sufficient conditions for a functor $F : C \to D$ of small categories to be a homotopy equivalence. Theorem B gives sufficient conditions to create homotopy fibration sequences from functors $F : C \to D$.

But first, we introduce the notion of a bisimplicial set and some of its properties.

Definition 4.11. A *bisimplicial set* X is a contravariant functor $\mathbf{\Delta} \times \mathbf{\Delta} \to$ **Sets**. Alternatively, it is a doubly indexed family $X_{p,q}$ of sets, together with *horizontal* face and degeneracy maps

$$\partial_i^h : X_{p,q} \to X_{p-1,q} \text{ and } \sigma_i^h : X_{p,q} \to X_{p+1,q}$$

and *vertical* face and degeneracy maps

$$\partial_i^v : X_{p,q} \to X_{p,q-1} \text{ and } \sigma_i^v : X_{p,q} \to X_{p,q+1}$$

such that the horizontal maps commute with the vertical maps, and both pairs follow the required relations among face and degeneracy maps as for simplicial sets. In particular, each $X_{\bullet,q}$ and $X_{p,\bullet}$ defines a simplicial set. See [BK72] for a detailed treatment of bisimplicial sets.

Definition 4.12. The *geometric realization* of a bisimplicial set X is the cellular complex BX given by taking the disjoint union of $X_{p,q} \times \Delta^p \times \Delta^q$ and identifying its horizontal and vertical faces with the appropriate $\Delta^{p-1} \times \Delta^q$ or $\Delta^p \times \Delta^{q-1}$.

Alternatively, it is the coend $\int_{p,q \in \mathbf{\Delta}} X_{p,q} \times \Delta^p \times \Delta^q$.

Note there is the *diagonal* functor from bisimplicial sets to simplicial sets $\operatorname{diag}(X) \mapsto X_{n,n}$, and it is well-known that BX is homeomorphic to $B \operatorname{diag}(X)$. See for instance, [BK72]. The following theorem is also well known: (see [Wal78])

Theorem 4.13. *Let $f : X \to Y$ be a map of simplicial sets.*

1. *If each simplicial map $X_{p,\bullet} \to Y_{p,\bullet}$ is a homotopy equivalence, so is $BX \to BY$.*
2. *If N is the nerve of a category C, though of as a bisimplicial set which is constant in the second coordinate, and $f^{-1}(c,_\bullet) \to f^{-1}(c',_\bullet)$ is a homotopy equivalence for every $c \to c'$ in C, then each $B(f^{-1}(c,_\bullet)) \to BX \to BC$ is a homotopy equivalence.*

Using this, we prove a lemma which eventually helps us prove Theorems A and B.

Lemma 4.14 (Quillen)**.** If $F : C \to D$ is a functor, the evident functor $D \backslash F \to C$ given by mapping $(d, d \to Fc)$ to c is a homotopy equivalence.

Proof. Let X denote the bisimplicial set where $X_{p,q}$ is the set of all pairs of sequences

$$(d_q \to \cdots \to d_0 \to F(c_0), c_0 \to \cdots \to c_p)$$

with the horizontal and vertical faces inherited from the nerves of C and D. Consider the projection $X \to NC$ which preserves only the second

sequence in the pair. Note that the discrete set NC_p is the set of all sequecnces $c_0 \to \cdots \to c_p$, and its inverse image is isomorphic to the nerve of $D\backslash F(c_0)$. $D\backslash F(c_0)$ is contractible since it has a terminal object $(c_0, \mathrm{id}_{F(c_0)})$. Using Theorem 4.13, from the homotopy fibration sequence we get $BX \xrightarrow{\simeq} BC$. The simplicial set $\mathrm{diag}(X)$ will be isomorphic to the nerve of $D\backslash F$, and thus we have the composite $B(D\backslash F) \xrightarrow{\simeq} BX \xrightarrow{\simeq} BC$, which concludes the proof.

□

Now, we are in a position to prove Theorem A.

Theorem 4.15 (Quillen's Theorem A)**.** *Let $F : C \to D$ be a functor of small categories. If the category $d\backslash F$ is contractible for every object d in D, then F is a homotopy equivalence.*

Proof. Consider the comma category $D\backslash F$ and the evident functors $D\backslash F \to C$ and $D\backslash F \to D^{op}$ given by mapping $(d, d \to Fc)$ to c and d respectively. Note that $B(D\backslash F) \to BC$ is a homotopy equivalence by 4.14.

We have a functor $D\backslash F \to D\backslash \mathrm{id}_D$ given by mapping $(d, d \to Fc)$ to (d, id_d), and a functor $D\backslash \mathrm{id}_D \to D^{op}$ given by mapping $(d, d \to Fc)$ to d. The latter is a homotopy equivalence since it is an equivalence of categories.

But we have a commutative diagram of categories

$$\begin{array}{ccccc} D^{op} & \longleftarrow & D\backslash F & \longrightarrow & C \\ \| & & \downarrow & & \downarrow \\ D^{op} & \longleftarrow & D\backslash \mathrm{id}_D & \longrightarrow & D \end{array}$$

So it suffices to show that $D\backslash F \to D^{op}$ is a homotopy equivalence. Let X be the bisimplicial set of Lemma 4.14. Then the map $B(D\backslash F) \xrightarrow{\pi} BD^{op}$ factors as $B(D\backslash F) \simeq BX \to BD^{op}$. The image of the simplicial map $\pi_{\bullet,q} : X_{\bullet,q} \to (ND^{op})_q$ is the discrete set of all sequences $d_q \to \cdots \to d_0$ in D. For any such sequence, the inverse image in $X_{\bullet,q}$ is isomorphic to the nerve of $d_0\backslash F$. Since we assumed that the latter to be contractible, by Theorem 4.13, we have the desired homotopy equivalence $B(D\backslash F) \xrightarrow{\simeq} BX \xrightarrow{\simeq} BD$. □

And then we have Theorem B, which will later be used to construct homotopy fibration sequences of categories.

Theorem 4.16 (Quillen's Theorem B)**.** *Let $F : C \to D$ be a functor of small categories such that every morphism $d \to d'$ in D induces a homotopy equivalence $d\backslash F \to d'\backslash F$. Then for any object d of D, the geometric*

realization of the sequence

$$d\backslash F \to C \to D$$

is a homotopy fibration sequence. Consequently, we have a long exact sequence

$$\cdots \to \pi_{n+1}(BD) \xrightarrow{\partial} \pi_n B(d\backslash F) \xrightarrow{j} \pi_n BC \xrightarrow{F} \pi_n BD \to \cdots$$

Proof. Consider the projection $\pi : X \to N(D^{op})$ as in Theorem A (4.15). Since the inverse image $\pi^{-1}(d)$ for any $d \in D$ is the nerve of $d\backslash F$ by the same argument, and we have the condition that $d \to d'$ induces $d\backslash F \to d'\backslash F$ is a homotopy euqivalence, we can use Theorem 4.13 to deduce that $B(d\backslash F) \to BX \to BD^{op}$ is a homotopy fibration sequence.

Since $BX \cong B(\operatorname{diag} X) \cong B(D\backslash F)$, we have $B(d\backslash F) \to B(D\backslash F) \to BD^{op}$ is a fibration sequence. Similarly, $d\backslash \operatorname{id}_D \to D\backslash \operatorname{id}_D \to D^{op}$ is also a fibration sequence. Consider the diagram

$$\begin{array}{ccccc} D^{op} & \longleftarrow & D\backslash F & \longleftarrow & d\backslash F \\ \| & & \downarrow{\scriptstyle F} & & \downarrow \\ D^{op} & \xleftarrow{\simeq} & D\backslash \operatorname{id}_D & \longleftarrow & d\backslash \operatorname{id}_D \end{array}$$

Note the $d\backslash \operatorname{id}_D$ is contractible. So $D\backslash \operatorname{id}_D \xrightarrow{\simeq} D^{op}$ is an equivalence, giving us the fibration sequence $B(d\backslash F) \to B(D\backslash F) \to B(D^{op}) = BD$. The theorem follows from the fact that $j : d\backslash F \to BC$ induces $B(D\backslash F) \xrightarrow{\simeq} BC$.

□

4.3 H-spaces

We will note that the following homotopy-theoretic generalization of a topological group will be relevant when considering upcoming constructions.

Definition 4.17 (H-space)**.** A H-space [2] X is a topological space with a binary operation $\mu : X \times X \to X$ and a distinguished element e such that

1. $\mu(e, e) = e$
2. $\mu(e, -) \simeq \mathrm{id}_X$
3. $\mu(-, e) \simeq \mathrm{id}_X$

Additionally, we say that a H-space is homotopy-associative or homotopy-commutative if those respective properties hold up to homotopy.

Topological groups are trivially examples of H-spaces. A class of illuminating nontrivial examples come from n-th loop spaces $\Omega^n X$ of some based space X, where the multiplication is given by concatenation of paths.

We refer the reader to [Whitehead] for basic facts about H-spaces, some of which we state.

Similar to group completions of monoids, it is a sensible question to ask for a universal way to complete a homotopy-commutative homotopy-associative H-space into having an *homotopy inverse*.

Suppose X is a homotopy-commutative homotopy-associative H-space. Then its set of connected components form an abelian monoid, and $H_0(X; \mathbb{Z})$ is the monoid ring $\mathbb{Z}[\pi_0(X)]$. Its integral homology $H_*(X; \mathbb{Z})$ is an associative graded-commutative ring via the Pontryagin product.

Definition 4.18. A *group-like* H-space X is a homotopy-associative homotopy-commutative with a homotopy inverse, that is a map $\nu : X \to X$ such that the composites

$$x \mapsto \mu(x, \nu(x)) \text{ and } x \mapsto \mu(\nu(x), x)$$

are homotopic to the constant map at e.

Theorem 4.19. *Let X be a homotopy-associative, homotopy-commutative H-space. A* group completion *of X is a H-space map $X \to Y$ such that $\pi_0(X) \to \pi_0(Y)$ is a group completion of the abelian monoid $\pi_0(X)$, and the natural map from the localization of the homology ring*

$$\pi_0(X)^{-1} H_*(X; k) \to H_*(Y; k)$$

is an isomorphism, for any commutative ring k.

[2] The H stands for Hopf

4.4 +-construction

The +-construction, devised by Kervaire, is a universal way to kill a perfect normal subgroup of the fundamental group of a connected CW complex, while preserving homology theory. The discussion of the +-construction fits into the framework of acyclic maps, which we explore here.

Definition 4.20. A space X is said to be acyclic if it has the homology of a point, i.e., $H_*(X;\mathbb{Z}) \cong H_*(\star;\mathbb{Z}) = \underline{\mathbb{Z}}$.

In particular, contractible spaces are acyclic. Acyclic spaces serve an analogous purpose to homology as contractible spaces do to homotopy.

Definition 4.21 (Acyclic maps)**.** Let $f : X \to Y$ be a map of based connected CW complexes. We say f is acyclic if the homotopy fiber $F(f)$ is acyclic, i.e., has the homology of a point.

In particular, a homotopy equivalence is acyclic since the former implies that the homotopy fiber is contractible. Note an acyclic map induces isomorphisms on integral homology $f_* : H_*(X;\mathbb{Z}) \cong H_*(Y;\mathbb{Z})$. Furthermore, we have the following result characterising acyclic maps:

Lemma 4.22. Let X, Y be based connected CW complexes. A map $f : X \to Y$ is acyclic if and only if for every $\pi_1(Y)$-module M, it induces an isomorphism $H_*(X;M) \cong H_*(Y;M)$.

The proof for this uses Serre Spectral sequences, which we skip and refer the reader to [**Weibel**].

Definition 4.23 (relative +-construction)**.** Let X be a based connected CW complex and P be a perfect normal subgroup of $\pi_1(X)$. Then we say a based acyclic map of CW-complexes $f : X \to Y$ is a +-construction relative to P if P is the kernel of the induced map $\pi_1(X) \to \pi_1(Y)$.

Since f was assumed to be acyclic, we can deduce from the exact sequence $\pi_1 F(f) \to \pi_1(X) \to \pi_1(Y) \to \pi_0 F(f)$ that the map $\pi_1(X) \to \pi_1(Y)$ is onto. (because $\pi_0 F(f) = \{\star\}$) And $\pi_1(Y) \cong \pi_1(X)/P$.

Theorem 4.24 (Quillen)**.** *Let X be a based connected CW complex and P be a perfect normal subgroup of $\pi_1(X)$.*

1. *There exists a +-construction $X \to Y$ relative to P.*
2. *A +-construction $f : X \to Y$ is universal among maps which kill P in the fundamental group. That is, for any $g : X \to Z$ such that the image of P in $\pi_1(Z)$ vanishes, we have a map $h : Y \to Z$, unique up to homotopy, such that $g = hf$.*

3. *In particular, if g is another +-construction relative to P, then there is a homotopy equivalence $h : Y \xrightarrow{\sim} Z$ compatible with the maps.*

The proof of this uses topological obstruction theory. We describe a construction for 1 in §4.4.1 and refer the reader to [Ber82] for a full proof of 1 and 2. 3 follows immediately from 2 since we have maps both way which must compose to identity up to homotopy.

Remark 4.4.1. Any group G has a maximal perfect normal subgroup, which we call the *perfect radical*. In fact, it is the subgroup generated by all perfect normal subgroups, hence their maximum. From now on, for any based CW complex X, we will denote by X^+ any +-construction on X relative to the perfect radical $P \triangleleft \pi_1(X)$.

4.4.1 Explicit Construction

A number of explicit *models* exist for the +-construction. Here, we present a construction of X^+ by attaching 2-cells to kill the maximal perfect subgroup of $\pi_1(X)$, and then 3-cells to recover the integral homology of X, making sure that the inclusion $X \hookrightarrow X^+$ is acyclic. We refer the reader to [Ger71] and [**Suslin**] for a variety of constructions. We will describe more models for K-theory in Sections §3 and §4.

Construction

Let X be a connected CW complex with a perfect normal subgroup P of the fundamental group $\pi_1(X)$. Identify the 1-cells in X corresponding to of P as e^1_p for each $p \in P$. Now construct a pushout Y by attaching disks D^2_p with the boundary S^2_p along every e^1_p. Call these new cells e^2_p.

That is, X' is the pushout of the diagram

$$\begin{array}{ccc} \coprod_{p\in P} S^1_p & \hookrightarrow & \coprod_{p\in P} D^2_p \\ \downarrow & & \downarrow \\ X & \hookrightarrow & Y \end{array}$$

Using the van-Kampen Theorem, we have the pushout through π_1 in **Grp**, which gives us $\pi_1(Y) = \pi_1(X) *_P 1 = \pi_1(X)/P$. (amalgamated product)

Consequently, H_1 is the abelianization. Since P is perfect, it is killed by abelianization and we have $H_1(Y) = H_1(X)$.

Consider the pair (Y, X). Since only 2-cells were attached, the homology of the pair is

$$H_2(Y, X) = \bigoplus_{[e_p^2]} \mathbb{Z}$$

and 0 elsewhere. We wish to cleverly attach 3-cells to kill relative homology without affecting H_3.

Consider the long exact sequence of the pair (Y, X)

$$\cdots \to H_{n+1}(Y, X) \xrightarrow{\partial} H_n(X) \to H_n(Y) \to H_n(Y, X) \xrightarrow{\partial} \cdots$$

At $n = 1$, since the map $H_1(X) \to H_1(Y)$ is an isomorphism, we have that $H_2(Y, X) \to H_1(X)$ is 0. Consequently, the map $H_2(Y) \to H_2(Y, X)$ is onto. We can lift each representative in $H_2(Y, X)$ to a representative in $H_2(Y)$, say σ_p for each class $[e_p^2]$.

Note each σ_p is represented by a map $h_p : S_p^2 \to Y$. Now attach 3-cells e_p^3 corresponding to each σ_p along the maps $h_p : S_p^2 \to Y$.

This yields another pushout which we call Z

$$Z = Y \cup \bigcup_{[e_p^2]} e_p^3$$

The triple $X \subseteq Y \subseteq Z$ induces a long exact sequence in relative homology

$$\cdots \to H_{n+1}(Z, Y) \xrightarrow{\partial} H_n(Y, X) \to H_n(Z, X) \to H_n(Z, Y) \xrightarrow{\partial} \cdots$$

At $n = 1$, since $H_2(Y, X) = 0$ and $H_0(Z, Y) = 0$, this gives us

$$\cdots \xrightarrow{\partial} 0 \to H_1(Z, X) \xrightarrow{\cong} H_1(Z, Y) \xrightarrow{\partial} 0 \to \cdots$$

Similarly, for $n \geq 4$, since $H_{n+1}(Z, Y) = 0$ and $H_{n-1}(Y, X) = 0$, we get

$$\cdots \xrightarrow{\partial} 0 \to H_n(Z, X) \xrightarrow{\cong} H_n(Z, Y) \xrightarrow{\partial} 0 \to \cdots$$

So we conclude $H_n(Z, X) \cong H_n(Z, Y) = 0$ for $n \neq 1, 2, 3$.

The only nontrivial terms in the sequence that remain are

$$\cdots \xrightarrow{\partial} 0 \to H_3(Z, X) \to H_3(Z, Y) \xrightarrow{\partial} H_2(Y, X) \to 0 \to \cdots$$

But $H_3(Z, Y)$ and $H_2(Y, X)$ are computable precisely due to construction. Z was constructed so as to kill

$$H_3(Z, Y) = \bigoplus_{[\sigma_p]} \mathbb{Z}$$

which is isomorphic via the evident maps to $H_2(Y, X) = \bigoplus_{[e_p]} \mathbb{Z}$.

So finally, we have $H_3(Z, X) = 0$, and altogether $H_n(Z, X) = 0$ for all n. Consequently, in the long exact sequence of the pair (Z, X), we get

$$\cdots \to H_{n+1}(Z, X) \xrightarrow{\partial} H_n(X) \to H_n(Z) \to H_n(Z, X) \xrightarrow{\partial} \cdots$$

which gives us that $H_n(X) \cong H_n(Z)$ induced via the inclusions are isomorphisms for all n.

It is clear that P is the kernel of the induced map $\pi_1(X) \to \pi_1(Y)$. This concludes the result that the stated construction preserves integral homology.

If this result were to extend to local coefficient systems, this would mean that the map $X \to Z$ is acyclic and forms a model for X^+. We do not pursue the full proof here and instead refer the reader to [Ros94]. Indeed, there is an alternate result in Proposition 5.6 which shows that what we have proven suffices for the case we are interested in, $BGL(R)^+$, which is what we shall discuss in the next chapter.

Chapter 5

Higher K-Theory and the K_n Groups

In this chapter, we shall explore the definitions of higher K-theory. The primary goal of higher K-theory is to (functorially) define a 'K-Theory space' $K(R)$ associated to a unital ring R. We then define the higher K-groups to be $K_n(R) = \pi_n K(R)$, and verify that this description matches with the classical theory for $n = 0, 1, 2$.

A number of approaches can be taken in different contexts, and we have some results showing that in some restricted contexts, these various constructions are equivalent, i.e., yield homotopy-equivalent K-theory spaces.

Finally, this allows us to define and relate the higher K-theory of finitely generated projective modules over a ring, among other applications.

In Section §1, we describe the $BGL(R)^+$ construction to define higher K-groups of a ring. In Section §2, we describe the $S^{-1}S$ construction on a symmetric monoidal category, and show its equivalence with the $+$-construction for $(\mathbf{P}(R), \oplus)$. In Section §3, we describe Quillen's Q-construction for exact categories. Section §4 is devoted to the $+ = Q$ theorem, which implies equivalence of the $+$ and Q-construction for $\mathbf{P}(R)$.

5.1 The $BGL(R)^+$ construction for a ring R

We apply the $+$-construction on the classifying space $BGL(R)$ of the stable general linear group over R. The maximal perfect subgroup of $GL(R) = \pi_1 BGL(R)$ is the stable group of elementary matrices, $E(R)$.

Theorem 5.1. *With this definition of $BGL(R)^+$, we obtain:*

1. *$\pi_1 BGL(R)^+ \cong K_1(R)$ with the kernel of the induced map $GL(R) =$*

$\pi_1 BGL(R) \to \pi_1 BGL(R)^+$ *being* $E(R)$.
2. *We have induced isomoprhisms* $H_*(BGL(R); M) \cong H_*(BGL(R)^+; M)$ *for every* $K_1(R)$*-module* M.

Proof. Since $E(R)$ is the maximal perfect normal subgroup of $GL(R)$, 1 follows immediately from Theorem 4.24. 2 follows from the characterization of acyclic maps in Lemma 4.22. □

Due to Theorem 4.24, $BGL^+(R)$ is uniquely defined up to homotopy equivalence via a canonical equivalence. Consequently, we may define $K_n(R) = \pi_n BGL(R)^+$ for $n \geq 1$. It is obvious that $K_1(R)$ defined here matches the description in Chapter 3. It remains to verify that this agrees with the classical theory for $n = 2$.

Proposition 5.2. Let P be a perfect normal subgroup of G with a $+$-construction $f : BG \to BG^+$ relative to P. Let $F(f)$ be the homotopy fiber of f. Then $\pi_1 F(f)$ is the universal central extension of P, and consequently $\pi_2(BG^+) \cong H_2(P; \mathbb{Z})$.

Proof. We have the fibration long exact sequence

$$\cdots \to \pi_2(BG) \to \pi_2(BG^+) \to \pi_1 F(f) \to G \to G/P \to 1$$

Since BG is an Eilenberg-Maclane space $K(G, 1)$, we have $\pi_2(BG) = 0$. So $\pi_2(BG^+)$ injects into $\pi_1 F(f)$, which in turns surjects onto the kernel of $G \to G/P$ which is P. Since $\pi_2(BG^+)$ is abelian, it must inject into the center of $\pi_1 F(f)$. This tells us that $\pi_1 F(f) \to P$ is a central extension. Since f is acylic, we have $\pi_1 F(f)$ is perfect and so $H_2(\pi_1 F(f), \mathbb{Z}) = 0$. By the Recognition Theorem of , we have that $\pi_1 F(f)$ is the universal central extension of P. Consequently, the kernel $\pi_2(BG^+)$ is given by $H_2(P; \mathbb{Z})$.

□

Corollary 5.3. We have $\pi_2(BGL(R))^+ \cong H_2(E(R); \mathbb{Z})$, and thus our definition matches with the classical theory since $H_2(E(R); \mathbb{Z}) \cong K_2(R)$ from Chapter 3.

However, at this point, it is not clear that the associations $R \mapsto BGL(R)^+$ or $R \mapsto K_n(R)$ are functorial. This invites some discussion.

5.1.1 Functoriality of the $BGL(R)^+$-construction

Apart from the upcoming models via $S^{-1}S$ and Q-constructions, we can make the following argument to show that the constructions of $BGL(R)^+$ can be performed functorially.

We can first define $BGL(\mathbb{Z})^+$ and take a homotopy pushout (using the double mapping cylinder) via the canonical map $\mathbb{Z} \to R$ which induces $BGL(\mathbb{Z}) \to BGL(R)$. Note that the homotopy pushout

$$\begin{array}{ccc} BGL(\mathbb{Z}) & \longrightarrow & BGL(\mathbb{Z})^+ \\ \downarrow & & \downarrow \\ BGL(R) & \longrightarrow & BGL(R)^+ \end{array}$$

is a plus construction for $BGL(R)$ and is functorial in $\mathbb{Z} \to R$.

5.1.2 H-Space Properties of $BGL(R)^+$

It's not hard to see that we can put a H-space structure on $BGL(R)^+$.

Proposition 5.4. $BGL(R)^+$ is a homotopy-associative H-space.

There is a recognition criteria for +-constructions of $BGL(R)$ in terms of homology and H-spaces, due to Quillen. The proof again uses obstruction theory. See [Ger71].

Theorem 5.5. *The plus construction $i : BGL(R) \to BGL(R)^+$ is universal for maps into H-spaces. That is, for any map $f : BGL(R) \to H$ for some H-space H, we have a map $g : BGL(R)^+ \to H$ so that $f = gi$, such that g is unique up to weak homotopy, i.e. the induced maps $\pi_n(BGL(R)^+) \to \pi_n(H)$ are independent of g.*

Proposition 5.6. If further $f_* : H_*(BGL(R); \mathbb{Z}) \xrightarrow{\cong} H_*(H; \mathbb{Z})$ is an isomorphism, then f is acyclic and g is a homotopy equivance.

5.2 $S^{-1}S$ construction

For a symmetric monoidal category $(S, \square)$, the construction of Chapter 2 for K_0 may be extended to a K-theory space $K^{\square}(S)$ by constructing an appropriate group-completion of S with respect to the monoidal operation $\square$.

Note that the geometric realization of a symmetric monoidal category is a homotopy-associative homotopy-commutative H-space with the multiplication coming from the monoidal product $\square$. We wish to construct a symmetric monoidal category $S^{-1}S$ which describes a group completion of BS so as to make $BS^{-1}S$ be a group-like H-space. [Gra76]

But first, for either of BS or $BS^{-1}S$ to be interesting, we must reduce our scope.

Definition 5.7 (symmetric monoidal groupoid)**.** A *symmetric monoidal groupoid* is a symmetric monoidal category $(S, \square)$ in which all morphisms are isomorphisms.

Further, we say S is *faithful* if all translations $s\square : \mathrm{Aut}(t) \to \mathrm{Aut}(s\square t)$ are injective.

Definition 5.8. Given a symmetric monoidal category S, $\mathrm{iso}(S)$ is the symmetric monoidal groupoid which is a subcategory of S consisting of the same objects but inheriting only isomorphisms.

A variety of symmetric monoidal groupoids yield interesting geometric realizations.

Example 5.2.1. The category $\mathbf{F}(R)$ with $\square = \oplus$ is contractible since it has an initial object 0. However $\mathrm{iso}(\mathbf{F}(R))$ is more interesting and in fact equivalent to the disjoint union $\coprod_n GL_n(R)$ where n runs through the integers.

Example 5.2.2. Similarly, 0 is also an initial object for the category $\mathbf{P}(R)$ with $\square = \oplus$. But $\mathrm{iso}(\mathbf{P}(R))$ is equivalent to $\coprod_P \mathrm{Aut}(P)$ where P runs over representatives of isomorphism clases of finitely generated projective modules.

Let S be a faithful symmetric monoidal groupoid.

Definition 5.9 ($S^{-1}S$)**.** Define $S^{-1}S$ to be a category where the objects are pairs of objects (m, n) for $m, n \in S$. Morphisms in $S^{-1}S$ from $(m_1, m_2) \to (n_1, n_2)$ are equivalence classes of triples (s, f_1, f_2) for some object $s \in S$ and $(f_1, f_2) : (s\square m_1, s\square m_2) \to (n_1, n_2)$ being a morphism in $S \times S$.

That is,

$$(m_1, m_2) \xrightarrow{s\square} (s\square m_1, s\square m_2) \xrightarrow{(f_1, f_2)} (n_1, n_2)$$

which we consider equivalent to

$$(m_1, m_2) \xrightarrow{t\square} (t\square m_1, t\square m_2) \xrightarrow{(g_1, g_2)} (n_1, n_2)$$

if there is an isomorphism $\alpha : s \xrightarrow{\cong} t$ such that $\alpha\square g_1 = f_1$ and $\alpha\square g_2 = f_2$. The composition $[(s, f_1, f_2)] \circ [(t, g_1, g_2)]$ is defined as $[(s\square t, f_1 \circ g_1, f_2 \circ g_2)]$. One may verify this is well-defined and indeed forms a category.

Remark 5.2.1. Consider two distinguished classes of morphisms in $S^{-1}S$. One is simultaneous translations, which we formally write $s\square : (m_1, m_2) \to (s\square m_1, s\square m_2)$, represented by $(s, \mathrm{id}, \mathrm{id})$. The other being pairs of maps $(f_1, f_2) : (m_1, m_2) \to (n_1, n_2)$ coming from morphisms in $S \times S$ where the translation composite is trivial. That is, the morphism is represented by (e, f_1, f_2).

Proposition 5.10. Let S be a faithful symmetric monoidal groupoid. Then any morphism $[(s, f_1, f_2)]$ in $S^{-1}S$ determines s, f_1, f_2 uniquely up to unique isomorphisms.

Proof. It is clear s, f_1, f_2 are determined uniquely up to isomorphism. Suppose (s, f_1, f_2) is equivalent to (t, g_1, g_2) via isomorphisms $\alpha, \beta : s \xrightarrow{\cong} t$ Since $\alpha \square g_i = f_i = \beta \square g_i$, we have $\alpha \square m_i = \beta \square m_i$ and since S was faithful, we recover $\alpha = \beta$. □

Thus, we can think of morphisms $[(s, f_1, f_2)]$ in $S^{-1}S$ as comprising of a *translation component* $[s\square]$ and *mapping component* $[(f_1, f_2)]$.

Remark 5.2.2. $S^{-1}S$ is a symmetric monoidal category with the multiplication given by $(m_1, m_2)\square(n_1, n_2) := (m_1 \square n_1, m_2 \square n_2)$. It is clear that the functor $S \to S^{-1}S$, which sends $s \mapsto (s, e)$ is a monoidal functor. Further, a monoidal functor $S \to T$ describes a (monoidal) natural transformation

$$\begin{array}{ccc} S & \longrightarrow & T \\ \downarrow & & \downarrow \\ S^{-1}S & \longrightarrow & T^{-1}T \end{array}$$

We also have an induced H-space map $BS \to B(S^{-1}S)$, and a map $\pi_0(S) \to \pi_0(S^{-1}S)$ of abelian monoids. (The component of s maps to the component of (s, e))

Further, $\pi_0(S^{-1}S)$ is an abelian group since for the connected coponent of (m, n) we have we have inverse (n, m) due to the isomorphism $(e, e) \xrightarrow{\cong} (m, n)\square(n, m)$ given by the mapping component $[(c_{mn}, c_{mn}^{-1})]$ where c_{mn} is the symmetric braiding isomorphism $m\square n \cong n\square m$ in S. It is abelian due to S being symmetric.

Definition 5.11. Let S be a symmetric monoidal groupoid. We define

$$K^{\square}(S) := B(S^{-1}S)$$

and consequently we can define $K_n^{\square}(S) = \pi_n B(S^{-1}S)$.

Note that a (strict) monoidal functor $S \to T$ induces $S^{-1}S \to T^{-1}T$ which in turn induces $K^{\square}(S) \to K^{\square}(T)$, Thus the $S^{-1}S$ construction is functorial!

Proposition 5.12. The group $K_0^{\square}(S) = \pi_0(S^{-1}S)$ is indeed the group completion of the monoid given by isomorphism classes of S via the inclusion $i_* : \pi_0(S) \to \pi_0(S^{-1}S)$.

Proof. Let $\varphi : \pi_0(S) \to G$ be a group completion of $\pi_0(S)$. We will show that the universal homomorphism $\widetilde{i}_* : G \to \pi_0(S^{-1}S)$ is invertible. Let j be the set map $S^{-1}S \to G$ as $j(m,n) := \varphi(m) - \varphi(n)$. If we have $s \in S$ then $j(m_1, m_2) = j(s\Box m_1, s\Box m_2)$ and if $f_i : m_i \to n_i$, then $j(m_1, m_2) = \varphi(m_1) - \varphi(m_2) = \varphi(n_1) - \varphi(n_2) = j(n_1, n_2)$. This shows j induces a well-defined homomorphism $j_* : \pi_0(S^{-1}S) \to G$. By construction, it is the inverse to $G \to \pi_0(S^{-1}S)$. □

This shows our definition matches the definition of $K_0(S)$ in Chapter 2.

5.2.1 Actions on other Categories

Definition 5.13. A monoidal category S is said to *act upon* a category X by a functor $\boxtimes : S \times X \to X$, if there are natural isomorphisms

$$s\Box(t \boxtimes x) \cong (s\Box t) \boxtimes x \qquad \text{and} \qquad e \boxtimes x \cong x$$

for $s, t \in S$ and $x \in X$, satisfying the coherence conditions for associativity and unit law, i.e. pentagonal and triangular diagrams of **??**.

For instance, S acts on itself by $\Box$. If X is a discrete category (a set), then S acts on X exactly when the monoid of isomorphism classes of S acts on X.

Definition 5.14 (Translation category)**.** If S acts upon X, then the *translation category* $\langle S, X\rangle$ is the category with the same objects as X, and morphisms $x \to y$ are equivalence classes of pairs $(s, s \boxtimes x \to y)$ for some $s \in S$.

Two pairs $(s, s\boxtimes x \to y)$ and $(t, t\boxtimes x \to y)$ are considered equivalent if there is an isomorphism $s \cong t$ such that the composite morphism $t \boxtimes x \cong s \to y$ is the same as $t \boxtimes x \to y$.

Example 5.2.3. Given a category X such that S acts on X by $\boxtimes$, we will write $S^{-1}X$ for the category $\langle S, S \times X\rangle$ with the action given by $s\boxtimes(t,x) = (s\Box t, s \boxtimes x)$. Note that if $X = S$, then this recovers $S^{-1}S$.

Remark 5.2.3. If S were a monoidal groupoid (every arrow in S is an isomorphism), then e would be an inital object of $\langle S, S\rangle$ and therefore we have that $S^{-1}\mathbf{1} \simeq B\langle S, S\rangle$ is contractible.

Definition 5.15 (invertible action)**.** We say that S acts *invertibly* if each translation $s\boxtimes : X \to X$ is a homotopy equivalence.

If S acts invertibly on $S^{-1}X$ by $s \boxtimes (t,x) = (s\Box t, x)$, which the homotopy inverse $s\boxtimes(t,x) = (t, s\boxtimes x)$ since there is a natural transformation $(t,x) \mapsto (s\Box t, s \boxtimes x)$.

We combine these obesrvations to get the following,

Proposition 5.16. Since $\pi_0 S$ is a multiplicatively closed subset of the ring $H_0(S) = \mathbb{Z}[\pi_0 S]$, it acts on $H_*(X)$ and acts invertibly on $H_*(S^{-1}X)$. The functor $X \to S^{-1}X$ described by $x \mapsto (0, x)$ induces the map

$$(\pi_0 S)^{-1} H_q(X) \to H_q(S^{-1}X) \tag{5.1}$$

We have the following result due to Quillen, showing us that $B(S^{-1}S)$ is indeed a group completion of BS. (See [Gra76])

Theorem 5.17 (Quillen)**.** *Let S be a faithful symmetric monoidal groupoid. Then (5.1) is an isomorphism for all X and q. In particular, $B(S^{-1}S)$ is a group completion of the H-space BS.*

5.2.2 Relation to the +-construction

In order to deduce $K(\mathbf{P}(R))$, we will first deduce $K(\mathbf{F}(R))$ and use a fact about cofinality to get our result.

Theorem 5.18 (Quillen)**.** *Let $S = \coprod GL_n(R)$ which is equivalent to $\mathbf{F}(R)$. Then $B(S^{-1}S)$ is the group-completion of BS and*

$$B(S^{-1}S) \simeq \mathbb{Z} \times BGL(R)^+$$

Proof. We will try to find an acyclic map from $BGL(R)$ to the connected basepoint component of $B(S^{-1}S)$. We perform this via the mapping telescope construction which describes the homotopy colimit.

$$BGL(R) = \varinjlim \operatorname{hocolim} BGL_n(R)$$

Homotopy classes of maps $BGL(R) \to B(S^{-1}S)$ correspond to compatible homotopy classes maps $BGL_n(R) \to B(S^{-1}S)$ for all n. Note that group maps $GL_n(R) \to \mathrm{Aut}_{S^{-1}S}(R^n, R^n)$ describe maps from $BGL_n(R) \to B(S^{-1}S)$. In fact, we have maps $f_n(g) = (g, 1)$ for each n and the following diagram commutes.

$$\begin{array}{ccc} GL_n(R) & \xrightarrow{f_n} & \mathrm{Aut}(R^n, R^n) \\ {\scriptstyle \square R}\downarrow & & \downarrow{\scriptstyle \square(RR)} \\ GL_{n+1}(R) & \xrightarrow{f_{n+1}} & \mathrm{Aut}(R^{n+1}, R^{n+1}) \end{array}$$

This implies we have natural transformations $f_n \Rightarrow f_{n+1}(\square R)$, which induce the homotopies $f_n \Rightarrow f_{n+1}(\square R) : BGL_n(R) \to B(S^{-1}S)$. Due to this compatibility of f_n, we get a map out of the mapping telescope $f : BGL(R) \to B(S^{-1}S)$.

Furthermore, the map lands in the connected component Y_S of the identity in $B(S^{-1}S)$, which is also a H-space.

We will now show that this map $BGL(R) \to Y_S$ is an isomorphism on integral homology. By Proposition 5.6, this will induce a homotopy equivalence $BGL(R)^+ \to Y_S$.

Let $e \in \pi_0 BS$ denote the path connected component of R in BS. By Theorem 5.17, $H_*(BS^{-1}S)$ is the localization of the ring $H_*(BS)$ at $\pi_0 S = \{e^n\}$. But this is just the colimit

$$H_*(BS) \to H_*(BS) \to H_*(BS) \to H_*(BS) \to \cdots$$

where the maps are given by the translation $\oplus R : S \to S$. So $H_* B(S^{-1}S) \cong H_*(Y_S) \otimes \mathbb{Z}[e, e^{-1}]$ which will allow us to recover $B(S^{-1}S) \simeq \mathbb{Z} \times Y_S$. Finally, we have the chain

$$H_*(Y_S) \cong \operatorname{colim} H_*(BGL_n(R)) = H_*(BGL(R))$$

And this shows $H_*(BGL(R)) \to Y_S$ is a homology isomorphism. □

5.2.3 Cofinality and Equivalence with +-construction

A monoidal functor $f : S \to T$ of faithful symmetric monoidal groupoids is said to be *cofinal* if for every t in T, there is some s in S and u in T such that $t \square u \cong f(s)$. The inclusion functor $\mathbf{F}(R) \to \mathbf{P}(R)$ for instance, is cofinal, because every projective module is a direct summand of a free module.

We have the following result which allows us to $K_n(T)$, a proof of which is outlined in [Wei13].

Theorem 5.19 (Cofinality Theorem)**.** *Let $f : S \to T$ be a cofinal monoidal functor of symmetric monoidal categories. Then*

1. *If T acts on X, then S acts on T via f, and $S^{-1}X \simeq S^{-1}X$.*
2. *If $\operatorname{Aut}_S(s) \cong \operatorname{Aut}_T(fs)$ for all s in S, then the basepoint components of $K(S)$ and $K(T)$ are homotopy equivalent. In particular, $K_n(S) \cong K_n(T)$ for $n \geq 1$.*

Owing to this Theorem, the fact that $\mathbf{F}(R)$ is cofinal in $\mathbf{P}(R)$, we have the following corollary.

Corollary 5.20. Let $S = \operatorname{iso} \mathbf{P}(R)$ be the (faithful) symmetric monoidal groupoid of finitely generated projective R-modules. Then

$$B(S^{-1}S) \simeq K_0(R) \times BGL(R)^+ \tag{5.2}$$

5.3 Quillen's Q-construction

In [Qui73], Quillen proposed another construction for the K-theory space. Given an exact category $\mathcal{A}$, we will define a category $Q\mathcal{A}$, and then define $K_n(\mathcal{A}) = \pi_{n+1}(BQ\mathcal{A})$ for all $n \geq 0$.

Definition 5.21 (Q-construction)**.** Let $\mathcal{A}$ be an exact category. We define $Q\mathcal{A}$ as the category with the same objects as $\mathcal{A}$, and the morphisms between $C \to C'$ are equivalence classes of diagrams

$$C \xleftarrow[j]{} D \xrightarrow{i} C'$$

where two such diagrams are considered isomorphic when there is an isomorphism of these diagrams which is identity on C and C'.

The composition of the above arrow with $C' \leftarrow D' \rightarrow C''$ is given by $C \leftarrow D \times_{C'} D' \rightarrow C''$ from the diagram

$$\begin{array}{ccccc} D \times_{C'} D' & \xrightarrow{\pi_2} & D' & \xrightarrow{i'} & C'' \\ {\scriptstyle \pi_1}\downarrow & & \downarrow{\scriptstyle j'} & & \\ D & \xrightarrow[i]{} & C' & & \\ {\scriptstyle j}\downarrow & & & & \\ C & & & & \end{array}$$

It is clear that composition is well-defined and associative and so $Q\mathcal{A}$ forms a well-defined category.

Remark 5.3.1. Admissible monomorphisms $C \rightarrowtail C'$ determine a morphism $C \to C'$ by taking $D = C'$ and $D \leftarrow C'$ as identity. We will call these morphisms *injective*.

Similarly, admissible epimorphisms $C \twoheadleftarrow C'$ determine a morphism $C \to C'$ by taking $D = C$ and $C \rightarrowtail D$ as identity. We will call these morphisms *surjective*.

Note that these distinguished families of morphisms are closed under composition, and generate all morphisms. Furthermore, any morphism $C \leftarrow D \rightarrowtail C'$ factors as $i_* j^*$ where i_* is the injective morphism determined by $D \rightarrowtail C'$ and j^* is the surjective morphism determined by $C \twoheadleftarrow D$. This factorization is unique up to unique isomorphism.

5.3.1 Alternative Description of Morphisms in $Q\mathcal{A}$

There is an alternative way to describe the morphisms $C \to D$ in $Q\mathcal{A}$ in terms of *subobjects* and *subquotients*. Fix some object C' in $\mathcal{A}$.

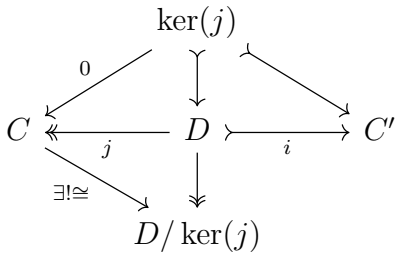
Definition 5.22. An *admissible subobject* of C' is an equivalence class of admissible monomorphisms $C \rightarrowtail C'$ with equivalence given by whether they factor through each other, or in other words as isomorphism in $\mathcal{A} \leftarrow C$.

An *admissible subquotient* is defined in the dual manner, using admissible epimorphisms.

One can define a partial order on the set of admissible monomorphisms into C'. We declare $C_1 \leq C_2$ if there is an admissible monomorphism $C_1 \rightarrowtail C_2$, which if it exists has to be unique. We then say (C_1, C_2) is an *admissible layer* in C' and the cokernel C_2/C_1 is an *admissible subquotient* of C'.

Proposition 5.23. The morphisms from $C \to C'$ may be identified with pairs $((C_1, C_2), \varphi)$, where (C_1, C_2) is an admissible layer in C' and $\varphi : C \xrightarrow{\cong} C_2/C_1$ is an isomorphism in $\mathcal{A}$.

Proof. For $C \twoheadleftarrow D \rightarrowtail C'$, take the admissible layer (C_1, C_2) where $C_2 = D$ and C_1 is the kernel of $C \twoheadleftarrow D$. Clearly, since C_1 gives a subobject of $D = C_2$ which gives a subobject of C', the given pair is indeed an admissible layer. A unique isomorphism follows since both C and the C_2/C_1 are cokernels of the kernel of an admissible epimorphism.



Conversely, an admissible layer (C_1, C_2) in C' evidently determines a morphism $C_2/C_1 \twoheadleftarrow C_2 \rightarrowtail C$ in $Q\mathcal{A}$ where the isomorphism may be taken as identity.

Furthermore, composition of morphisms $C \to C'$ and $C' \to C''$ may be described this way and arises from the fact that subquotient of a subquotient is a subquotient.

Let morphisms $C \to C'$ and $C' \to C''$ correspond to $((C_1, C_2), \varphi)$ and $((C_1', C_2'), \psi)$ where (C_1, C_2) and (C_1', C_2') are admissible layers in C and C'' respectively with isomorphisms $\varphi : C \xrightarrow{\cong} C_2/C_1$ and $\psi : C' \xrightarrow{\cong} C_2'/C_1'$. Let the map $C_2' \to C_2'/C_1'$ be denoted q.

Then the composite $C \to C' \to C''$ corresponds to the layer (D_1, D_2) where $D_1 = C_1 \times_{C'} C_2'$ and $D_2 = C_2 \times_{C'} C_2'$ with the isomorphism $\Phi : C \xrightarrow{\cong} D_2/D_1$. This follows from the evident isomorphisms and the fact that D_2 is the pullback of $C_2 \rightarrowtail C' \twoheadleftarrow C_2'$. $\square$

This characterization is particularly helpful to eludicate certain properties of the category $Q\mathcal{A}$.

Example 5.3.1. Using this characterization, we can describe the morphisms with source 0 in $Q\mathcal{A}$. Morphisms $0 \to C$ are in 1-1 correspondence with the subobjects of C, thus injective. There is a canonoical morphism $C \to 0$ since 0 has a unique subquotient.

Example 5.3.2. Isomorphisms $C \xrightarrow{\cong} D$ in $Q\mathcal{A}$ are in 1-1 correspondence with isomorphisms $C \xrightarrow{\cong} D$ in $\mathcal{A}$.

5.3.2 K-Theory using the Q-construction

We now prove a few homotopy-theoretic properties of the Q-construction, before we eventually define K_n-groups corresponding to a given (skeletally small) exact category.

Proposition 5.24. An exact functor $F : \mathcal{A} \to \mathcal{B}$ induces a functor $QF : Q\mathcal{A} \to Q\mathcal{B}$.

Proof. On the level of objects, it suffices to define QF precisely as F. Suppose $g : C \to D$ is a morphism in $Q\mathcal{A}$. We may identify g with an admissible layer (D_1, D_2) of D. Define $QF(g)$ to be the morphism $FC \to FD$ in $Q\mathcal{B}$ corresponding to the admissible layer (FD_1, FD_2) of FD, which is indeed an admissible layer since the exactness relations in $\mathcal{A}$ carry over to $\mathcal{B}$ by the exact functor F. Functoriality follows since compositions are defined naturally across $Q\mathcal{A}$ and $Q\mathcal{B}$. □

We present a proof of the theorem proven by Quillen in [Qui73].

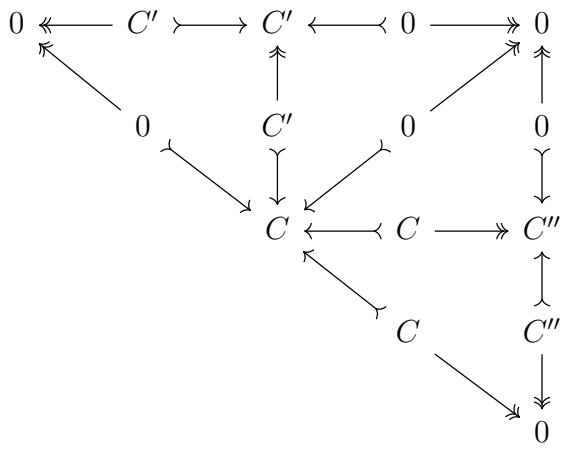
Theorem 5.25 (Quillen)**.** *Let $\mathcal{A}$ be a small exact category. The fundmental group of $BQ\mathcal{A}$ at the basepoint 0 is canonically isomorphic to the $K_0(\mathcal{A})$ group defined in Chapter 2.*

Proof. Note that $\pi_1(BQ\mathcal{A})$ is generated by loops given by the morphisms $0 \to C \leftarrow 0$ for each C in $Q\mathcal{A}$, where $0 \to C$ is the injective morphism given by the subobject $0 \rightarrowtail C$, and $C \leftarrow 0$ is the surjective morphism given by $0 \twoheadleftarrow C$. This also shows $BQ\mathcal{A}$ is (path) connected.

It is clear that the loops given by $0 \to C \leftarrow 0$ determine C up to isomorphism, and loops $0 \to C \leftarrow 0$ generate $\pi_1(BQ\mathcal{A})$.

Given any short exact sequence $C' \rightarrowtail C \twoheadrightarrow C''$ in $\mathcal{A}$, we wish to show that the loop given by the concatenated diagram $0 \to C' \leftarrow 0 \to C'' \leftarrow 0$ in $Q\mathcal{A}$

is homotopic to the loop given by $0 \to C \leftarrow 0$. Drawing out the morphisms of these diagrams in $Q\mathcal{A}$, we have the following diagram in $\mathcal{A}$



Note that there are four triangles with vertices and morphisms in $Q\mathcal{A}$ which divide this diagram as displayed here. To show the desired claim, it suffices to show that each of the four traingles commute in $Q\mathcal{A}$, which would imply each corresponds to a 2-simplex in $\pi_1(BQ\mathcal{A})$, making the loop $0 \to C' \leftarrow 0 \to C'' \leftarrow 0$ homotopic to $0 \to C \leftarrow 0$ due to shared edges.

But in each of the triangles the computation of the pullback in the composition is trivial and yields either the injective morphism $0 \to C$ (by $0 \twoheadleftarrow 0 \rightarrowtail C$) or the surjective morphism $0 \to C$ (by $0 \twoheadleftarrow C \rightarrowtail C$). In any case, it is equal to the composite edge.

It also follows that $\pi_1(BQ\mathcal{A})$ is abelian.

We can now define a group homomorphism $\Theta : K_0(\mathcal{A}) \to \pi_1(BQ\mathcal{A})$ by mapping $[C]$ to the loop $0 \to C \leftarrow 0$ as mentioned. This is a (surjective) group homomorphism since exact sequences lead to homotopic loops.

□

Definition 5.26. Let $\mathcal{A}$ be a small exact category. Then $K\mathcal{A}$ is defined to be the space $\Omega BQ\mathcal{A}$ pointed at 0 and further

$$K_n(\mathcal{A}) := \pi_n(K\mathcal{A}) = \pi_{n+1}(BQ\mathcal{A})$$

This definition agrees with $K_0(\mathcal{A})$ defined in Chapter 2. Further, any exact functor $F : \mathcal{A} \to \mathcal{B}$ induces a functor $Q\mathcal{A} \to Q\mathcal{B}$, hence $BQ\mathcal{A} \to BQ\mathcal{B}$ showing that $K_n(\mathcal{A}) \to K_n(\mathcal{B})$ showing that the Q-construction leads to a functorial K-theory.

Definition 5.27. If R is a unital ring, we define $K(R) := K(\mathbf{P}(R))$ and consequently

$$K_n(R) = K_n(\mathbf{P}(R)).$$

This matches our original definition for $n = 0$. We prove in §5.4 the $+ = Q$ Theorem, which implies that this matches with the description of the K_n groups defined using the $+$-construction in §5.1.

Definition 5.28. If R is a Noetherian ring with unity, we define $G(R) := K(\mathbf{M}(R))$ and consequently

$$G_n(R) := K_n(\mathbf{M}(R)).$$

This gives us G-Theory of unital Noetherian rings and matches the classical definition for $n = 0$.

5.3.3 Key Properties of the Q-construction

Morita Invariance

Since K-theory from the Q-construction is functorial, for Morita-equivalent rings R, S, i.e., there is an equivalence $\mathbf{P}(R) \simeq \mathbf{P}(S)$ of categories, from which it follows that $K_n(R) \cong K_n(S)$.

5.4 The "$+ = Q$" Theorem

Given an additive category $\mathcal{A}$, there are two approaches to defining K-theory over $\mathcal{A}$. One is through

Theorem 5.29 (Quillen)**.** *Let $\mathcal{A}$ be a split-exact category and $S = \operatorname{iso}\mathcal{A}$ be the symmetric monoidal groupoid of isomorphisms in $\mathcal{A}$ with $\square = \oplus$, then*

$$\Omega BQ\mathcal{A} \simeq B(S^{-1}S)$$

Corollary 5.30 ($+ = Q$)**.** For any unital ring R,

$$\Omega BQ\mathbf{P}(R) \simeq K_0(R) \times BGL(R)^+$$

This means that the K-Theory of $\mathbf{P}(R)$ from the Q-construction is equivalent to the K-Theory defined using the $+$-construction.

5.4.1 Proof of the $+ = Q$ Theorem

For the proof, we wish to produce a fibration sequence

$$S^{-1}S \to \mathbf{??} \to Q\mathcal{A}$$

where **??** is chosen appropriately and $B(\mathbf{??})$ is contractible. Then upon taking the homotopy long exact sequence of the fibration, we would have our desired homotopy equivalence $\Omega BQ\mathcal{A} \simeq BS^{-1}S$. This sequence of leading up to the theorem is taken from [Wei13].

Definition 5.31. Given an exact category $\mathcal{A}$, we define $\mathcal{E}\mathcal{A}$ where the objects of $\mathcal{E}\mathcal{A}$ are admissible exact sequences in $\mathcal{A}$, and a morphism between sequences $C' \rightarrowtail C \twoheadrightarrow C''$ and $D' \rightarrowtail D \twoheadrightarrow D''$ is an equivalence class of the following diagrams admssible exact sequences $D' \rightarrowtail C \twoheadrightarrow E$ such that the following diagram commutes

$$\begin{array}{ccccc} C' & \rightarrowtail & C & \twoheadrightarrow & C'' \\ {\scriptstyle\alpha}\uparrow & & \| & & \uparrow \\ D' & \rightarrowtail & C & \twoheadrightarrow & E \\ \| & & {\scriptstyle\beta}\downarrow & & \downarrow \\ D' & \rightarrowtail & D & \twoheadrightarrow & D'' \end{array}$$

Two such diagrams are considered equivalent when there is an isomorphism between them which is identity at all vertices except for E.

Note since the right edge in the diagram describes a morphism in $Q\mathcal{A}$, we have a functor $f : \mathcal{E}\mathcal{A} \to Q\mathcal{A}$ given by picking C'' from the exact sequence $C' \rightarrowtail C \twoheadrightarrow C''$.

We now consider the fiber category $f^{-1}(E)$ for an object E which we denote $\mathcal{E}_E$.

Remark 5.4.1. If we consider the identity morphism to be fixed at E , then the fiber category $\mathcal{E}_E$ consists of exact sequences with target E, and the morphisms from $C' \rightarrowtail C \twoheadrightarrow E$ to $D' \rightarrowtail D \twoheadrightarrow E$ are diagrams of the form

$$\begin{array}{ccccc} C' & \rightarrowtail & C & \twoheadrightarrow & E \\ {\scriptstyle\alpha}\uparrow{\scriptstyle\cong} & & {\scriptstyle\beta}\downarrow{\scriptstyle\cong} & & \| \\ D' & \rightarrowtail & D & \twoheadrightarrow & E \end{array}$$

It follows automatically that each morphism in $\mathcal{E}_E$ is an isomorphism which we may describe as a pair (α, β) where $\alpha : D' \cong C'$ and $\beta : C \cong D$.

The following result allows us to relate the fiber category with the symmetric monoidal groupoid $S = \operatorname{iso}\mathcal{A}$.

Proposition 5.32. The fiber category $\mathcal{E}_0$ is homotopy equivalent to $S = \operatorname{iso}\mathcal{A}$.

Proof. Let $F : S \to \mathcal{E}_0$ be the functor given by mapping an object C to $C \rightarrowtail C \twoheadrightarrow 0$, and an isomorphism $f : C \to C$ to the morphism (f, f^{-1}) with the diagram

$$\begin{array}{ccccc} C & \rightarrowtail & C & \twoheadrightarrow & E \\ {\scriptstyle f}\uparrow & & {\scriptstyle f^{-1}}\downarrow & & \| \\ C & \rightarrowtail & C & \twoheadrightarrow & E \end{array}$$

In fact, since every object in $\mathcal{E}_0$ is naturally isomorphic to such a sequence, the functor F is defined naturally. It is also a full embedding. This gives us the claim: □

Proposition 5.33. For any object E in $\mathcal{A}$, $\mathcal{E}_E$ is a symmetric monoidal category, and there is a (faithful) monoidal functor $\eta_E : S \to \mathcal{E}_E$ sending C to the sequence $C \rightarrowtail C \oplus E \twoheadrightarrow E$.

Proof. We define the monoidal structure on $\mathcal{E}_E$ as follows. Let $e = (0 \rightarrowtail E \twoheadrightarrow E)$ be the identity. Let x denote $C' \rightarrowtail C \twoheadrightarrow E$ and y denote $D' \rightarrowtail D \twoheadrightarrow E$. Then we define $x \square y$ as

$$C' \oplus D' \rightarrowtail C \times_E D \twoheadrightarrow E$$

It follows that this obeys the necessary unit, commutatitivity and coherence laws and does define a symmetric monoidal structure on $\mathcal{E}_E$. The symmetric monoidal functor $S \to \mathcal{E}_E$ follows naturally by mapping C to the split-exact sequence $C \rightarrowtail C \oplus E \twoheadrightarrow E$. It is also faithful. □

Now, here is a wonderful observation which reduces the complexity of the fiber category $\mathcal{E}_E$ in a restricted case.

Remark 5.4.2. If $\mathcal{A}$ were split-exact, then every sequence in $\mathcal{E}_E$ is of the form $C \rightarrowtail C \oplus E \twoheadrightarrow E$, hence comes from S via the functor η_E. Consequently, the category $\langle S, \mathcal{E}_E \rangle$ is connected.

Proposition 5.34. If $\mathcal{A}$ is split exact, each $S^{-1}S \to S^{-1}\mathcal{E}_E$ is a homotopy equivalence.

Proof. Note the sequence $S^{-1}S \to S^{-1}\mathcal{E}_E \to \langle S, \mathcal{E}_E \rangle$ is a fibration. It suffices to show that $L = \langle S, \mathcal{E}_E \rangle$ is contractible. Note that the monoidal operation on $\mathcal{E}_E$ induces a monoidal operation on L, making BL a H-space. Since L is connected, by [Whi78] it implies BL is group-like and has a homotopy inverse.

Now for any exact sequence $x = (C \rightarrowtail C \oplus E \twoheadrightarrow E)$ we have a diagonal morphism $\Delta_x : x \to x \square x$ in L,

$$\begin{array}{ccccc} C & \rightarrowtail & D & \twoheadrightarrow & C \\ \downarrow & & \downarrow & & \| \\ C \oplus C & \rightarrowtail & D \times_C D & \twoheadrightarrow & C \end{array}$$

The diagonal Δ_x is natural in x. This natural transformation induces a homotopy between $\mathrm{id} : BL \to BL$ and $(-)^2 : BL \to BL$. Using the homotopy inverse to subtract away id, we get $\mathrm{id} : BL \to BL$ is null-homotopic.

Hence BL is contractible. □

Lemma 5.35. For any morphism $f : E' \to E$ in $Q\mathcal{A}$, there is a canonical "base-change" functor $f^* : \mathcal{E}_E \to \mathcal{E}_{E'}$ and a natural transformation $\eta : f^* \Rightarrow i$, where i is the inclusion $\mathcal{E}_E \hookrightarrow \mathcal{E}\mathcal{A}$.

Theorem 5.36. *If $\mathcal{A}$ is a split exact category and $S = \operatorname{iso}\mathcal{A}$, then the sequence*

$$S^{-1}S \to S^{-1}\mathcal{E}\mathcal{A} \to Q\mathcal{A}$$

is a homotopy fibration.

Proof. We use Quillen's Theorem B to prove this. For Theorem B to apply, we need to show that the base changes f^* are homotopy equivalences.

We show the same on properly injective and properly surjective morphisms $0 \to C$. For properly injective morphisms, the base change is given by $(C \rightarrowtail D \twoheadrightarrow E) \mapsto (C \rightarrowtail C \twoheadrightarrow 0)$ which can be identified with C in S. The composition $S^{-1}S \to S^{-1}\mathcal{E}_E \to Q\mathcal{A}$ is thus identity.

For properly surjective morphisms, the base change is $(C \rightarrowtail D \twoheadrightarrow E) \mapsto (D \rightarrowtail D \twoheadrightarrow -1)$ which can be identified with D in S. The composition $S^{-1}S \to S^{-1}\mathcal{E}_E \to Q\mathcal{A}$ in this case is $C \mapsto (D = C \oplus E)$, which is realized by the natural transformation given by $E\oplus$, hence is also a homotopy equivalence.

Hence we can conclude that in general, f^* is a homotopy equivalence. □

Proof of Theorem 5.29. The proof follows from Theorem 5.36 if we show that $S^{-1}\mathcal{E}\mathcal{A}$ is contractible. Note that $\mathcal{E}\mathcal{A}$ is contractible. Any action of S on a contractible category must be invertible. So we have the homotopy equivalence $\mathcal{E}\mathcal{A} \to S^{-1}\mathcal{E}\mathcal{A}$, concluding that $S^{-1}\mathcal{E}\mathcal{A}$ is contractible. □

This concludes the section and gives us Theorem 5.29 and Corollary 5.30.

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